Effects of different financial frictions on households*

Francesco Ferlaino †

November 2023

Abstract

This paper aims to study how financial frictions of different typologies affect households' wealth and consumption after a contractionary monetary policy shock, and whether these differences are significant. The two types of frictions considered for this purpose are frictions on production firms and frictions on households' borrowing ability, both of which are embedded in a HANK model. The results suggest that the friction in the productive sector affects wealth inequality to a greater extent, while the friction on household loans induces a higher dispersion of consumption compared to the counterfactual scenario. The two dynamics appear to be related. In the model with frictions on household borrowings, households are discouraged from moving to the bottom of the distribution, reducing loans and therefore consumption capacity. On the other hand, when an active friction is present on the production side of the economy, more households are pushed to the bottom. This fluctuation relatively increases the Gini index of wealth, but simultaneously allows greater economy-wide consumption smoothing, resulting in less consumption inequality when compared to the previous case.

1 Introduction

Although more than a decade has passed since the burst of the real estate bubble, the effects of the Great Financial Crisis remain tangible in economic research. Two branches of literature have drawn particular attention from academics interested in macro models to better understand the causes and effects of such events.

One concerns the implications of considering heterogeneous households as opposed to standard New Keynesian (NK) models, where a Representative Agent (RA) exists. This change in perspective is mostly motivated by the rising inequality (in terms of both wealth and income) experienced not only in the United States, but also in almost all advanced

*I am grateful to my advisor, Domenico Delli Gatti, for his invaluable support and guidance. I particularly thank Alice Albonico, Diego Bohórquez, Pietro Bomprezzi, Jacopo Bonchi, Andrea Colciago, Davide Debortoli, Marco Membretti, Francesco Menoncin, Hugo Minnella, Maria Laura Parisi, Giuseppe Spataro and Barna Szabó for helpful discussions and comments. Any remaining errors are my own.

†DEMS, Università degli studi di Milano - Bicocca. Email: f.ferlaino@campus.unimib.it.

economies. Although this phenomenon has been ongoing for more than 40 years,[1] the 2008 financial crisis exacerbated this process. RA models have the great advantage of being potentially solvable analytically; therefore, they are not time-consuming when solved with computer software, a feature that in the past was particularly important because of the lower computing power. Still, the RA assumption is an extreme simplification that becomes even more stringent when a greater share of households is "Hand-to-Mouth" (HtM), that is, when they simply consume their income with almost no ability to save, a feature typical of rising inequality. To account for that, more complex models are required. A good trade-off is represented by Two-Agents New Keynesian (TANK) models: two types of households are present here, HtM and non-HtM households, keeping the model fairly simple but with important implications for aggregate dynamics and monetary policy transmission mechanisms. Nonetheless, these types of models do not allow for an analysis of changes in the shape of the household distribution of wealth, whereas in full-fledged Heterogeneous Agents New Keynesian (HANK) models, several households act differently, both in terms of consumption and saving behavior. These models allow for the matching of wealth distribution moments, which turn out to be relevant for explaining the consequences of aggregate shocks.

The other area of interest involves considering amplification mechanisms in theoretical models that could justify large variable fluctuations, even after an aggregate shock of a moderate size. Another common assumption in standard Real Business Cycle (RBC) and NK models is that the economy is frictionless: financial intermediaries are almost nonexistent or simply transfer liquidity without impediments, and funds always land in the hands of agents who can maximize their returns. Although these relatively simple models have proven to be good at approximating historical business cycle statistics, they are often unable to justify the magnitude and persistence of aggregate shock effects. The assumption of perfectly functioning financial markets, convenient as it may be, is, again, not realistic, particularly in times of financial crisis and credit rationing. Theories of imperfect financial markets began to circulate almost a hundred years ago (e.g., Fisher, 1933) in the aftermath of another disruptive financial crisis, the 1929 stock market crash. Models that could explain such dynamics have been developed since the 1970s (e.g., Akerlof, 1970), but they probably did not receive enough attention, given that, until the Great Recession, there was widespread consensus that financial crises were something of the past or, if anything, a problem concerning primarily underdeveloped economies. Since 2008, an increasing number of scholars have reconsidered the importance of financial frictions, integrating existing mechanisms, and creating new model features that could better explain how a relatively small disturbance could lead to violent and persistent effects.

[1] Piketty (2017) provides a thorough review of the recent inequality history, especially in advanced economies.

There is also a growing body of literature on the effects of monetary policy shocks on heterogeneous households. Coibion et al. (2017) is probably one of the most influential empirical contribution. Using data on consumption and income from the Consumer Expenditure Survey (CEX), the authors prove that contractionary monetary policies seem to have significant persistent effects, leading to higher levels of income, labor earnings, consumption, and total expenditure inequality. From a modeling perspective, the seminal paper by Kaplan et al. (2018) is probably one of the most important in demonstrating how monetary policy transmission mechanisms act very differently in a full-fledged HANK model compared to the relative RANK and TANK versions. However, to date, few studies have dealt with monetary shock effects on household behavior in an environment with financial frictions.

This study aims to verify whether a conventional contractionary monetary policy shock (i.e., a rise in the nominal interest rate by the central bank) has different effects on household distributions of wealth and consumption, depending on the type of financial friction considered in the model. I consider two types of frictions: frictions on the ability of production firms to raise external funds, and frictions on the ability of households to obtain loans for consumption smoothing. In both cases, the friction magnitude is proportional to the spread between the rate at stake (gross return on capital in the case of frictions on firms, loan rate for frictions on households) and the risk-free rate. This means that when these spreads increase, a financial accelerator is generated, which enhances the effects of the aggregate shock. The empirical results of recent studies show a positive correlation between these spreads and inequality indices, particularly consumption dispersion measures.[2] Lee et al. (2020) provide empirical evidences on the relationship between consumption dispersion and two measures of the spread concerning household borrowings: the spread between the two year personal loan rate and the three months T-Bill rate and the spread between the Commercial Bank interest on three months Credit Card plans and the three months T-Bill rate. Using Local Projection regression á la Jordà (2005), they show that an increase in the spread is associated with higher consumption inequality, regardless of the interest rate spread considered. Following a similar methodology, Ferlaino (2023) finds that a similar relationship occurs between an increase in consumption inequality and the credit spread for non-financial firms, employing the "GZ credit spread" built by Gilchrist and Zakrajšek (2012).

Starting from these premises, I build a full-fledged HANK model featuring asset market incompleteness, idiosyncratic income risk, sticky prices, and two potential sources of financial frictions that come into play, depending on the case. I want to emphasize that my intent is not to give accurate quantitative results since the asset heterogeneity in the

[2]Consumption inequality is often used as a preferred measure in the empirical literature, since time series from the CEX have quarterly data for the US. Wealth dynamics are more difficult to extrapolate, because the Survey of Consumer Finances is triennial.

model is monodimensional.[3] Instead, I conduct a qualitative analysis to address the direction in which household distributions tend to move in the aftermath of a contractionary monetary shock in a particular state of the economy. Regarding frictions on productive firms, I resort to a financial accelerator similar to that proposed by Bernanke et al. (1999), which is one of the most seminal and recurring in the financial friction literature. In the case of friction on household borrowing ability, I partially follow Cúrdia and Woodford (2016) and assume that the spread between the deposit and loan interest rate is proportional to a non-decreasing convex function of the aggregate level of household debt in the economy:[4] a higher amount of household debt causes an increase in the spread. As the mechanisms and complexities of the two financial accelerators differ, the magnitude of the impulse responses could be different, but not necessarily because of economic dynamics. Therefore, for comparison reasons, I do not apply the same monetary shock for the two cases, but I use two different magnitudes that generate similar fluctuations for output.

The main result is that the type of friction is important for changes in savings and consumption. The firms' financial accelerator generates a relatively higher increase in wealth inequality, while financial frictions on households induce relatively higher consumption dispersion.[5] This behavior is due to how the two frictions affect the distribution around the zero-wealth threshold, which sorts households between borrowers and savers. When I consider only frictions on the productive side of the economy, leaving the consumer loan spread rate fixed, more households become borrowers, because borrowing is relatively cheaper. This implies a rise in the Gini index of wealth inequality: households are pushed to the bottom of the distribution, where they become borrowers and can better smooth their consumption because of the fixed loan spread. By contrast, when only frictions on households' borrowing ability are in place, households are discouraged from borrowing by the increasing borrowing penalty. Therefore, a greater share of households choose to become HtM, consuming just what they get from their income, increasing consumption dispersion but relatively decreasing wealth inequality, since fewer households decide to move to the bottom of the distribution. This translates into a different aggregate composite consumption response between the two models, whereas other variables have similar impulse responses in most cases.

This paper touches on different fields of macroeconomics. First, the model has roots in the literature concerning high heterogeneity among households, a path that started at the end of the 1980s (Imrohoroğlu, 1989; Huggett, 1993; Aiyagari, 1994). Krusell and Smith (1998) were the first to include aggregate uncertainty in such models, although their

[3] Accurate quantitative results are best suited for HANK models featuring two different assets (usually liquid and illiquid) as in Kaplan et al. (2018) or Luetticke (2021), since only with this kind of structure it is possible to target "wealthy hand-to-mouth" households.

[4] Cúrdia and Woodford (2016) also assume that the spread could be a function of households defaulting on unsecured debts. I neglect this feature to keep the model as simple as possible.

[5] Both wealth and consumption inequality are calculated as Gini indices.

findings suggest that household heterogeneity has little impact on aggregate dynamics, such as consumption, investment, and output.However, the Great Recession and increase in inequalities have raised interest in this topic, and an increasing number of scholars are striving to build algorithms capable of solving models with greater and more complex heterogeneity (e.g., Bayer and Luetticke, 2020; Auclert et al., 2021).

Second, this paper fits within the branch that examines the implications of household heterogeneity for monetary policy. Kaplan et al. (2018) is one of the most important contributions to the field, proving that household heterogeneity is fundamental in understanding monetary policy transmission. Nonetheless, concerns about monetary policy mechanisms with household heterogeneity were at the center of a blooming body of literature in recent years (e.g., Auclert, 2019; Luetticke, 2021). A thorough survey on this topic can be found in Colciago et al. (2019).

Third, my findings contribute to expanding the vast literature on financial frictions. Most frictions are built around the concept of asymmetric information between the lender-household and borrower-firm (e.g., Akerlof, 1970; Stiglitz and Weiss, 1981; Bernanke et al., 1999). Another common feature is the idea that a "moral hazard" exists that prevents the credit market from being frictionless (e.g., Holmstrom and Tirole, 1997; Gertler and Karadi, 2011; Farhi and Tirole, 2012). Papers on household borrowing frictions usually focus on unsecured loans and credit tightening (e.g., Iacoviello, 2005; Chatterjee et al., 2007; Cúrdia and Woodford, 2016). The survey by Brunnermeier et al. (2012) provides an excellent summary of the state-of-the-art in this branch of the economic literature.

By contrast, the theoretical literature on the effects of aggregate shocks on heterogeneous households in an environment featuring financial frictions is probably at the beginning of his existence. Guerrieri and Lorenzoni (2017) focuses on frictions on households and study the effects of a credit crunch on consumer spending by applying two types of shocks: a narrowing of the borrowing constraint and an increase of the borrowing spread for households. Nakajima and Ríos-Rull (2019) study the effects of earning shocks in a model with unsecured consumer credit and consumer bankruptcies. Fernández-Villaverde et al. (2023) analyze the nonlinear linkages between aggregate and financial variables in a framework featuring household heterogeneity and a leveraged "financial expert". In terms of monetary policy evaluation, Lee et al. (2020) investigate how an increase in the nominal risk-free rate (among other aggregate shocks) affects aggregate variables and household distributions of wealth and consumption when the household borrowing rate is affected by financial intermediaries' moral hazard. Ferlaino (2023) focuses on the impact on heterogeneous households of a financial accelerator generated by a leveraged production sector.

Differently from what this study seeks to accomplish, none of the papers cited above inspect whether movements in inequality measures could depend on the types of frictions

considered in the economy and, if that is the case, what is the reason behind the different responses.

The remainder of this paper is organized as follows. Section 2 outlines the model. Section 3 depicts the solution method for the model. Section 4 explains the calibration strategy. Section 5 displays results. Section 6 gives summary conclusions.

2 The model

To obtain a better comparison between the two financial frictions, I do not compare two different models (one for each friction). Instead, I build a model incorporating both frictions so that the starting point for the analysis (i.e., the steady-state) is the same. I then turn on one friction or the other and compare the impulse responses.[6] The model comprises households, financial intermediaries, a production sector, a central bank, and the government. Households consume, earn income (from either labor or profit, according to their household type), save, and borrow in a liquid asset. This asset yields an interest rate, that is augmented by a borrowing penalty in case of loans. There are two types of financial intermediaries: commercial banks, which intermediate household borrowings, and investment banks, which intermediate firm borrowings. The production sector produces goods and capital. The central bank is in charge of monetary policy and sets the nominal interest rate, whereas the government acts as fiscal authority and chooses how to finance government spending. The behavior of each agent is explained in detail below.

2.1 Households

There is a continuum of ex-ante identical households of measure one indexed by $i \in [0, 1]$. They are infinitely lived, have time-separable preferences with a time discount factor β, and derive felicity from consumption, c_{it}, and leisure.

Following Bayer et al. (2019), I assume households have Greenwood–Hercowitz–Huffman (GHH) preferences (Greenwood et al., 1988) and maximize the discounted sum of felicity:

$$V = E_0 \max_{\{c_{it}, l_{it}\}} \sum_{t=0}^{\infty} \beta^t u\left(c_{it} - G(h_{it}, l_{it})\right) \ . \tag{1}$$

Assuming GHH preferences instead of separable preferences has two major advantages and one flaw. First, from a computational perspective, it simplifies the numerical analysis. Second, as explained by Auclert et al. (2023), this prevents the model from generating an excessive Marginal Propensity to Earns (MPE), especially for households with high

[6] Because the two frictions are based on an interest rate spread fluctuating endogenously, the "active" friction, depending on the case, is the one for which the spread varies over time. To "shut down" a friction, I fix the relative spread.

Marginal Propensity to consume (MPC), since GHH preferences dampen wealth effects on labor supply. However, using GHH preferences in models with household heterogeneity translates into higher fiscal and monetary multipliers. The latter should not be a problem in this model because both the scenarios compared in this analysis would be affected by this issue.[7]

The felicity function features Constant Relative Risk Aversion (CRRA):

$$u(x_{it}) = \frac{x_{it}^{1-\xi}}{1-\xi} \ , \tag{2}$$

where $\xi \geq 0$ is the risk-aversion parameter, and $x_{it} = (c_{it} - G(h_{it}, l_{it}))$ is household i's composite demand for goods consumption and leisure. The function G measures the disutility from work.

Goods consumption bundles differentiated goods j according to a Dixit–Stiglitz aggregator:

$$c_{it} = \left(\int c_{ijt}^{\frac{\eta-1}{\eta}} dj \right)^{\frac{\eta}{\eta-1}} \ . \tag{3}$$

Each of these differentiated goods is offered at price p_{jt}, so that for the aggregate price level, $P_t = \left(\int p_{jt}^{1-\eta} dj \right)^{\frac{1}{1-\eta}}$, the demand for each of the varieties is given by:

$$c_{ijt} = \left(\frac{p_{jt}}{P_t} \right)^{-\eta} c_{it} \ . \tag{4}$$

The disutility of work, $G(h_{it}, l_{it})$, determines a household's labor supply given the aggregate wage rate, W_t , and a labor income tax, τ, through the first-order condition:

$$\frac{\partial G(h_{it}, l_{it})}{\partial l_{it}} = (1-\tau) W_t h_{it} \ . \tag{5}$$

Assuming that G has a constant elasticity with respect to labor, I can write:

$$\frac{\partial G(h_{it}, l_{it})}{\partial l_{it}} = (1+\gamma) \frac{G(h_{it}, l_{it})}{l_{it}} \ , \tag{6}$$

with $\gamma > 0$ being the Frisch elasticity of labor supply. The expression of the composite good can be simplified, making use of (5) and (6):

$$x_{it} = c_{it} - G(h_{it}, l_{it}) = c_{it} - \frac{(1-\tau) W_t h_{it} l_{it}}{1+\gamma} \ . \tag{7}$$

[7] Auclert et al. (2023) call it the "New Keynesian models trilemma". Regarding HANK models, choosing separable preferences delivers consistent MPC and multipliers but not MPE. On the other hand, choosing GHH preferences delivers consistent MPC and MPE but not multipliers. One solution proposed by the authors is to consider a HANK model with separable preferences and sticky wages. As interesting as it may be as a model integration, I believe that for the time being, such a complication is not necessary.

Since the Frisch elasticity of labor supply is a constant parameter, the disutility of labor is always a constant fraction of labor income. Therefore, in both the household budget constraint and its felicity function, only after-tax income enters, and neither hours worked nor productivity appears separately. This implies that, as suggested by Bayer et al. (2019), it can be assumed that $G(h_{it}, l_{it}) = h_{it}\frac{l_{it}^{1+\gamma}}{1+\gamma}$ without further loss of generality, as long as we treat the empirical distribution of income as a calibration target. This functional form simplifies the household problem, as h_{it} drops out from the first-order condition, and all households supply the same number of hours $l_{it} = L(W_t)$. Total effective labor input, $\int l_{it}h_{it}di$, is therefore equal to $L(W_t)$ since $\int h_{it}di = 1$. [8]

There are two types of household: workers and rentiers. Workers supply labor, L_t, in the production sector and have positive idiosyncratic labor productivity, $h_{it} > 0$. Their income is $W_t h_{it} L_t$. Rentiers have zero labor productivity, $h_{it} = 0$, but collect a proportional share of total profits generated from the production sector, Π_t. Idiosyncratic labor productivity h_{it} follows an exogenous Markov chain according to the following first-order autoregressive process and a fixed probability of transition between the worker and rentier state:

$$h_{it} = \begin{cases} exp(\rho_h log(h_{it-1}) + \epsilon_{it}^h) & \text{with probability } 1-\zeta \text{ if } h_{it-1} \neq 0 \\ 1 & \text{with probability } \iota \text{ if } h_{it-1} = 0 \\ 0 & \text{else} \end{cases} \tag{8}$$

with $\epsilon_{it}^h \sim N(0, \sigma_h)$. The parameter $\zeta \in (0,1)$ is the probability that a worker becomes a rentier and $\iota \in (0,1)$ is the probability that a rentier becomes a worker. As stated above, workers that become rentiers leave the labor market ($h_{it} = 0$), while rentiers that become workers are endowed with median productivity ($h_{it} = 1$).[9] Workers and rentiers pay the same level of taxation,τ, on their income.

The asset market is incomplete: there are no Arrow-Debreu state-contingent securities; households self-insure themselves only through savings in a non-state contingent risk-free liquid asset, a_{it}, and they can borrow up to an exogenous borrowing limit. The household i budget constraint is:

$$c_{it} + a_{it+1} = \left(\frac{R_t^I}{\pi_t}\right) a_{it} + (1-\tau)(W_t h_{it} L_t + \mathbf{I}_{h_{it}=0}\Pi_t) \ , \qquad a_{it} \geq \underline{a} \ , \tag{9}$$

where $\mathbf{I}_{h_{it}=0}$ takes value 1 if household i is a rentier, or 0 otherwise. On the right-hand side, we have households' expenditure, that is, consumption, c_{it} and 1-year-maturity savings, a_{it+1}. The left-hand side corresponds to households' total earnings, that is, the

[8] More specifically, deriving the FOC with respect to labor of the households' optimization problem, making use of the new assumed $G(h_{it}, l_{it})$, and combining it with (5), we obtain $l_{it} = [(1-\tau)W_t]^{\frac{1}{\gamma}} = L_t$, since l_{it} depends only on aggregate variables.

[9] Appendix A contains details on the transition matrix for household productivity.

work/rent income net of taxes, $(1-\tau)(W_t h_{it} L_t + \mathbf{I}_{h_{it}=0}\Pi_t)$, plus earnings (expenses) from savings (borrowings) in the liquid asset, $\left(\frac{R_t^I}{\pi_t}\right) a_{it}$.

According to (7), total goods consumption can be expressed as $c_{it} = x_{it} + \frac{(1-\tau)W_t h_{it} l_{it}}{1+\gamma}$. By substituting this equation into (9), I can rewrite the household budget constraint in terms of composite consumption, x_{it}:

$$x_{it} + a_{it+1} = \left(\frac{R_t^I}{\pi_t}\right) a_{it} + (1-\tau)\left(\frac{\gamma}{1+\gamma} W_t h_{it} L_t + \mathbf{I}_{h_{it}=0}\Pi_t\right) , \qquad a_{it} \geq \underline{a} . \tag{10}$$

Equation (10) states that, in this model, what matters for households is the intertemporal allocation of composite consumption, x_{it}, rather than total goods consumption, c_{it}.

R_t^I is the gross nominal return on liquid assets. Borrowing households pay a "penalty", ω_t^H, on the interest rate when they ask for a loan. Therefore, R_t^I has two definitions based on household i's wealth:

$$R_t^I = \begin{cases} R_t & \text{if } a_{it} \geq 0 \\ R_t(1+\omega_t^H) & \text{if } a_{it} < 0 \end{cases} \tag{11}$$

The model tracks only net household financial positions. This means that households cannot save and borrow simultaneously. Aggregate liquidity, $A_t = \int a_{it} di$, comprises household savings, and borrowings, B_t. In turn, households can save in three types of deposits that yield the same interest rate: deposits directed to commercial banks and used for household loans, D_t^H, deposits directed to investment banks and used for firm loans, D_t^F, and government bonds, D_t^G. Therefore, I can write the aggregate level of liquidity in the hands of households as:

$$A_t = D_t^H + D_t^F + D_t^G - B_t . \tag{12}$$

Since these three saving instruments yield the same interest rate, households are completely indifferent to their portfolio composition.[10]

2.2 Financial intermediaries

Financial intermediaries collect deposits from households and promise returns equal to the risk-free interest rate. There are two types of intermediaries: commercial banks, which specialize in intermediations among households, and investment banks, which specialize

[10]For sake of simplicity, I assume that the portfolio composition of any saver household is the same, and equal to the aggregate level of the three saving instruments.

in intermediation between households and the production sector.[11] These two types of financial intermediaries define the different types of financial frictions introduced in the model. First, I explain how commercial banks act before moving to investment banks.

2.2.1 Commercial Banks - Financial frictions on households

Commercial banks act similarly to the financial intermediaries in Cúrdia and Woodford (2016). I assume that banks can lend at most an amount that suffices to allow them to repay what they own to their depositors, considering the higher loan rate that households must pay when borrowing. This implies:

$$R_t(1+\omega_t^H)B_t = R_t D_t^H \ . \tag{13}$$

Furthermore, when originating loans, commercial banks burn resources according to a non-decreasing, weakly convex function of the aggregate level of household debt, $\Xi_t(B_t)$. Therefore, end-of-the-period profits for commercial banks are:

$$\Pi_t^{com} = D_t^H - B_t - \Xi_t(B_t) \ . \tag{14}$$

Using (13), (14) can be rewritten as:

$$\Pi_t^{com} = \omega_t^H B_t - \Xi_t(B_t) \ . \tag{15}$$

Since commercial banks are in perfect competition, a bank chooses B_t that maximizes profits, leading to the F.O.C.:

$$\omega_t^H = \Xi_t'(B_t) \ , \tag{16}$$

with the function $\Xi_t(B_t) = \tilde{\Xi} B_t^{\eta^{FF}}$, with $\tilde{\Xi}$ and η^{FF} being calibrated parameters.

Result (16) directly links the penalty on household borrowings, ω_t^H, to the aggregate level of household debt.[12] An increase in household indebtedness economy-wide results in a higher borrowing penalty, causing further depression in economic activities.

2.2.2 Investment Banks - Financial frictions on firms

Investment banks collect deposits from households and promise returns equal to the real risk-free interest rate, R/π. For ease of display, I assume that the production sector is run

[11] This is obviously an abuse of terminology if we consider the real meaning of commercial and investment banks, but the idea is to give intuitive names to intermediaries that could clearly distinguish the functions of the two types of bank.

[12] It is worth noting that empirical data usually show a higher degree of indebtedness in richer households, who can borrow more and at lower rates using as collateral their accumulated wealth (which, most of the time, is illiquid). However, we should keep in mind that this is a relatively simple model of net financial positions; therefore, it is impossible (and out of the scope of this study) to take track of such dynamics.

by entrepreneurs, who are a mass-zero group of managers who are entitled to all the profits generated in the production sector and rebate them to rentier households. Investment banks and entrepreneurs are responsible for the other financial friction considered in this model. Following Bernanke et al. (1999), I assume a continuum of entrepreneurs, indexed by j. Entrepreneur j acquires capital, K_j, from capital producers at the end of period t which is used at time $t+1$. To buy capital for production, entrepreneurs rely on two type of financing: internal financing (equity), N_j, and external financing (debt), D_j^F, borrowed from investment banks.

Entrepreneur j's balance sheet at period $t+1$ is:

$$q_t K_{jt+1} = N_{jt+1} + D^F_{jt+1} \ , \tag{17}$$

where q is the price of capital during the purchasing period.

One prerequisite for the financial accelerator to work is that entrepreneurs are not indifferent to the composition of their balance sheets; that is, external financing is more expensive than internal financing. To do so, I introduce a "Costly State Verification" (CSV) problem à la Townsend (1979) in which lenders (investment banks) must pay a fixed auditing cost in order to observe the realized returns of borrowers (entrepreneurs). A relatively higher demand for debt increases auditing costs, resulting in a lower level of aggregate capital obtained for production.

Entrepreneurs repay investment banks with a portion of their realized returns on capital. In this framework, entrepreneurs are risk-neutral, whereas households are risk-averse. This implies a loan contract in which entrepreneurs absorb any aggregate risk on the realization of their profits. I also assume the existence of an idiosyncratic shock to entrepreneur j, ω_j^F,[13] on the gross return on aggregate capital, R^K. The idiosyncratic shock ω^F has a log normal distribution of mean $E(\omega^F) = 1$ that is i.i.d. across time and entrepreneurs, with a continuous and once differentiable c.d.f., $F(\omega^F)$.[14]

The optimal contract for investment banks is:

$$\bar{\omega}^F_{jt+1} R^K_{t+1} q_t K_{jt+1} = Z_{jt+1} D_{jt+1} \ , \tag{18}$$

where Z_j is the gross non-default loan rate and $\bar{\omega}_j^F$ is the threshold value for entrepreneur j such that, for $\omega^F_{jt+1} \geq \bar{\omega}^F_{jt+1}$, entrepreneur j repays $Z_{jt+1}D_{jt+1}$ to banks and retains $\omega^F_{jt+1} R^K_{t+1} q_t K_{jt+1} - Z_{jt+1} D_{jt+1}$. In the case of $\omega^F_{jt+1} < \bar{\omega}^F_{jt+1}$, instead, she cannot repay and defaults on her debt, obtaining nothing. Since entrepreneurs' future realizations of capital returns are only known by entrepreneurs ex-post, investment banks must pay a

[13] As noted by Christiano et al. (2014), ω^F could be thought of as the idiosyncratic risk in actual business ventures: in the hands of some entrepreneurs, a given amount of raw capital is a great success, while in other cases may be not.

[14] Appendix B.1 provides analytical expressions for $F(\omega^F)$ and other functions used in the following equations.

fixed auditing cost, μ, to recover what is left of entrepreneur j's activity after default, obtaining $(1-\mu)\omega^F_{jt+1}R^K_{t+1}q_tK_{jt+1}$.

Because of the optimal contract, investment banks should receive an expected return equal to the opportunity cost of their funds. By assumption, they hold a perfectly safe portfolio (i.e., they are able to perfectly diversify the idiosyncratic risk involved in lending), and the opportunity cost for investment banks is the real gross risk-free rate, R/π. It follows that the participation constraint for investment banks that must be satisfied in each period $t+1$ is:

$$[1-F(\bar{\omega}^F_{jt+1})]Z_{jt+1}D_{jt+1}+(1-\mu)\int_0^{\bar{\omega}^F_{jt+1}}\omega^F_j dF(\omega^F_j)R^K_{t+1}q_tK_{jt+1}\geq\frac{R_{t+1}}{\pi_{t+1}}D_{jt+1}\ , \quad (19)$$

where $F(\bar{\omega}^F_j)$ is entrepreneur j default probability. Since financial markets are in perfect competition, (19) must hold with equality. The first term on the left-hand side of (19) represents the revenues received by investment banks from the fraction of entrepreneurs that do not default, whereas the second term is what investment banks can collect from defaulting entrepreneurs after paying monitoring costs.

Following the notation proposed in Christiano et al. (2014), I combine (17), (18), and (19) to write the following relationship:

$$EFP_{jt+1}=f\left(\bar{\omega}^F_{jt+1},LEV_{jt+1}\right)\ ,\ \text{with}\ f'(LEV_{jt+1})>0\ . \quad (20)$$

where EFP is the "External Finance Premium" that Bernanke et al. (1999) define as the ratio between the return on capital and the real risk-free rate, $R^K/(R/\pi)$, and $LEV=qK/N$ is entrepreneur j's leverage. The EFP can be considered a measure of the cost of external funds for the entrepreneur and, therefore, as a proxy for the strength of financial frictions. The $\left(\bar{\omega}^F_{jt+1},LEV_{jt+1}\right)$ combinations that satisfy (20) define a menu of state $(t+1)$-contingent standard debt contracts offered to entrepreneur j, who chooses the contract that maximizes its objective.

In Appendix B.2, I illustrate the entrepreneur j's optimization problem, which provides three important outcomes. First, the EFP increases monotonically with LEV. This means that entrepreneurs with a higher level of leverage pay a higher EFP. Second, the threshold value for entrepreneur j's default, $\bar{\omega}^F_j$, is endogenously defined by the EFP. Third, the fact that $\bar{\omega}^F_j$ depends only on the aggregate variables (R, R^K and π) implies that every entrepreneur will choose the same firm structure, that is, $\bar{\omega}^F$ and LEV. Therefore, it is possible to drop superscript j in the notation and consider a representative entrepreneur.

The other fundamental equation for the functioning of this financial accelerator is the law of motion for entrepreneurs' equity, which is expressed as follows:

$$N_{t+1} = \gamma^F \left[q_{t-1} R_t^K K_t - \frac{R_t}{\pi_t} D_t - \mu G(\bar{\omega}_t^F) q_{t-1} R_t^K K_t \right] . \tag{21}$$

Equation (21) states that entrepreneurs' equity after the production process at time t is equal to the gross return on capital net of the loan repayment and auditing costs (which are borne by entrepreneurs because they are risk-neutral). Parameter γ^F represents the share of surviving entrepreneurs who bring their equity to the production process from one period to the next. Conversely, the share of entrepreneurs $1 - \gamma^F$ dies and consumes equity at time t (we can think of this as entrepreneurial consumption). As explained by Carlstrom et al. (2016), this assumption avoids excessive entrepreneurs' self-financing in the long run.

Note that in (21) I did not included entrepreneurial labor, as usual in the literature (e.g., Bernanke et al., 1999, Christiano et al., 2014). The assumption of entrepreneurial labor was introduced mainly to justify the initial amount of equity for new entrepreneurs that take the place of the dead ones. However, to keep the model as simple as possible, I follow Carlstrom et al. (2016), assuming that new entrepreneurs' initial equity comes from a lump-sum transfer from existing entrepreneurs. Even so, since the funding can be arbitrarily small and since only aggregate equity matters, this transfer can be neglected in equation (21).[15]

Alternatively, (21) can be written in a more compact form as:

$$N_{t+1} = \gamma^F \left[1 - \Gamma(\bar{\omega}_t^F) \right] R_t^K q_{t-1} K_t \ , \tag{22}$$

where $\left[1 - \Gamma(\bar{\omega}_t^F) \right]$ is the share of capital returns to which the non-defaulting entrepreneurs are entitled.[16] Equation (22), together with (20), explain this financial accelerator mechanism. Equation (20) states that an increase in entrepreneurs' leverage increases also the EFP. At the same time, (22) tells that an increase in the EFP increases $\bar{\omega}^F$ as well, negatively affecting entrepreneurs' equity level for the next period and, therefore, impacting the aggregate leverage.

2.3 Intermediate-goods producers

Intermediate-goods producers adopt a standard Cobb-Douglas production function with constant returns to scale, employing aggregate capital, K, supplied by entrepreneurs and labor, L, from workers:

$$Y_t = z_t L_t^{\alpha} K_t^{1-\alpha} \ , \tag{23}$$

[15] Bernanke et al. (1999) keep the share of income going to entrepreneurial labor at a very low level (on the order of 0.01), therefore neglecting this income sounds as a reasonable model simplification.

[16] See Appendix B.2

where z is the Total Factor Productivity (TFP).

TFP follows a first-order autoregressive process of type:

$$log(z_t) = \rho_z log(z_{t-1}) + \epsilon_t^z \ , \tag{24}$$

with ϵ_t^z following a normal distribution with mean 0 and variance σ^z.

Intermediate goods producers sell their production to resellers at a relative price MC_t. Therefore, their profit optimization is given by:

$$\Pi_t^{IG} = MC_t z_t L_t^{\alpha} K_t^{1-\alpha} - w_t L_t - r_t^K K_t \ . \tag{25}$$

Since they are in perfect competition, their profit optimization problem returns the wage paid per unit of labor and the rent paid per unit of capital:

$$W_t = \alpha MC_t z_t \left(\frac{K_t}{L_t}\right)^{(1-\alpha)} \ , \tag{26}$$

$$r_t^K = (1-\alpha)\, MC_t z_t \left(\frac{L_t}{K_t}\right)^{\alpha} \ . \tag{27}$$

2.4 Resellers

Resellers are agents assigned to differentiate intermediate goods and set prices. Price adjustment costs follow a Rotemberg (1982) setup, and resellers preserve entrepreneurial characteristics.[17] The demand for the differentiated good g is:

$$y_{gt} = \left(\frac{p_{gt}}{P_t}\right)^{-\eta} Y_t \ , \tag{28}$$

where $\eta > 1$ is the elasticity of substitution and p_g is the price at which good g is purchased.

Given (28) and the quadratic costs of price adjustment, the resellers maximize:

$$E_0 \sum_{t=0}^{\infty} \beta^t Y_t \left\{ \left(\frac{p_{gt}}{P_t} - MC_t\right) \left(\frac{p_{gt}}{P_t}\right)^{-\eta} - \frac{\eta}{2\kappa} \left(log \frac{p_{gt}}{p_{gt-1}}\right)^2 \right\} \ , \tag{29}$$

[17] Bayer et al. (2019) make the further assumption that price setting is delegated to a mass-zero group of households (managers) that are risk neutral and compensated by a share in profits. Since in my model the whole production sector is run by entrepreneurs that, by assumption, are risk neutral and entitled to all the profits generated in this sector, I do no need to make this further assumption.

with a time-constant discount factor.[18]

The New Keynesian Phillips Curve (NKPC) derived from the F.O.C. for price setting is as follows:

$$log(\pi_t) = \beta E_t \left[log(\pi_{t+1}) \frac{Y_{t+1}}{Y_t} \right] + \kappa \left(MC_t - \frac{\eta - 1}{\eta} \right) , \tag{30}$$

where π_t is the gross inflation rate defined as $\frac{P_t}{P_{t-1}}$.

2.5 Capital producers

After production at time t, entrepreneurs sell depreciated capital to capital producers at a price q_t. They refurbish depreciated capital at no cost,[19] and uses goods as investment inputs, I_t, to produce new capital, $\Delta K_{t+1} = K_{t+1} - K_t$, subject to quadratic adjustment costs. Finally, they resell the newly produced capital to entrepreneurs before entering the next period (therefore still at price q_t). The law of motion for capital producers is:

$$I_t = \Delta K_{t+1} + \frac{\phi}{2} \left(\frac{\Delta K_{t+1}}{K_t} \right)^2 K_t + \delta K_t \ . \tag{31}$$

where δ is the depreciation rate for capital.

Then, they maximize their profits, $q_t \Delta K_{t+1} - I_t$, w.r.t. newly produced capital, ΔK_{t+1}. This optimization problem delivers the optimal capital price:

$$q_t = 1 + \phi \frac{\Delta K_{t+1}}{K_t} \ . \tag{32}$$

Equation (32) ensures that if the level of aggregate capital increases over time, so does its price.

It follows that entrepreneurs' return on capital does not depend only on goods production, but also on fluctuations in capital price; since entrepreneurs buy capital at the end of the period, with the price of that period, they see their capital at the beginning of the next period appreciated (depreciated) if q increases (decreases). The gross return on capital employed at time t can be written as:

$$R_t^K q_{t-1} K_t = r_t^K K_t + q_t K_t (1 - \delta) \ , \tag{33}$$

where the first term on the right-hand side is the marginal productivity of capital de-

[18] As explained by Bayer et al. (2019), only the steady-state value of the discount factor matters in the resellers' problem, due to the fact that I calibrate to a zero inflation steady-state, the same value for the discount factor of managers and households and approximate the aggregate dynamics linearly. This assumption simplifies the notation, since fluctuations in stochastic discount factors are virtually irrelevant.

[19] The "no cost" assumption does not mean that δK is refurbished for free. Capital producers still need to buy the exact amount of I necessary to refurbish depreciated capital, but do not waste any further resources in this process. In fact, the law of motion for capital producers in the steady-state (when $\Delta K = 0$) is $I = \delta K$.

rived in (27), and the second term represents eventual capital gains (or losses) net of depreciation. I can rearrange and finally derive the gross interest rate of capital as:

$$R_t^K = \frac{r_t^K + q_t(1-\delta)}{q_{t-1}} \ . \tag{34}$$

2.6 Final-goods producers

Final-goods producers are perfectly competitive, buy differentiated goods from resellers at a given price, and produce a single homogeneous final good that is used for consumption, government spending, and investment. The optimization problem of final-goods producers is:

$$\max_{\{Y_t, y_{gt} \in [0,1]\}} P_t Y_t - \int_0^1 p_{gt} y_{gt} dg \ , \tag{35}$$

subject to the following Constant Elasticity of Substitution (CES) function:

$$Y_t = \left(\int_0^1 (y_{gt})^{\left(\frac{\eta-1}{\eta}\right)} dg \right)^{\left(\frac{\eta}{\eta-1}\right)} \ . \tag{36}$$

From the zero-profit condition, the price index of the final good is:

$$P_t = \left(\int_0^1 (p_{gt})^{(1-\eta)} \right)^{\left(\frac{1}{1-\eta}\right)} \ . \tag{37}$$

2.7 Central bank

The central bank is responsible for the monetary policy. It sets the gross nominal risk-free interest rate, R_t, reacting to the deviation from steady-state inflation, and engages interest rate smoothing. The Taylor-type rule employed by the central bank is as follows:

$$\frac{R_{t+1}}{\bar{R}} = \left(\frac{R_t}{\bar{R}}\right)^{\rho_R} \left(\frac{\pi_t}{\bar{\pi}}\right)^{(1-\rho_R)\rho_\pi} \epsilon_t^R \ , \tag{38}$$

where ϵ_t^R is the monetary policy shock defined as $log(\epsilon_t^R) \sim N(0, \sigma_R)$. The parameter $\rho_R \geq 0$ rules the interest rate smoothing (if $\rho_R = 0$, the next-period interest rate depends only on inflation), whereas ρ_π captures the magnitude of the central bank's response to inflation fluctuations: the larger ρ_π, the stronger the central bank reaction (for the case limit $\rho_\pi \to \infty$, the inflation is perfectly stabilized at its steady-state level).

2.8 Government

The government acts as fiscal authority. It determines the level of public expenditure, G_t, tax revenues, T_t and issuance of new bonds, D^G_{t+1}. Its budget constraint is given by:

$$D^G_{t+1} = \left(\frac{R_t}{\pi_t}\right) D^G_t + G_t - T_t \ , \tag{39}$$

where T_t are the taxes collected from both workers and rentier households:

$$T_t = \tau \left(W_t h_{it} L_t + \mathbf{I}_{h_{it}=0} \Pi_t\right) \ . \tag{40}$$

Bond issuance is regulated by the following rules:

$$\frac{D^G_{t+1}}{\bar{D}^G} = \left(\frac{D^G_t \frac{R_t}{\pi_t}}{\bar{D}^G \frac{\bar{R}}{\bar{\pi}}}\right)^{\rho_{gov}} \ . \tag{41}$$

Coefficient ρ_{gov} captures how fast the government wants to balance its budget. When $\rho_{gov} \to 0$, the government aims to balance its budget by adjusting spending. Instead, when $\rho_{gov} \to 1$, the government is willing to roll over most of the outstanding debt.

2.9 Market clearing

Let $\Theta_t(a, h)$ be the joint distribution of liquid assets and productivity across households on date t.

The liquid asset market clears when:

$$\int a^*(a, h)\Theta_t(a, h) da dh = A_t \ , \tag{42}$$

where $a^*(a, h)$ is the optimal saving policy function of the household.

The market for capital clears for (31) and (32).

The labor market clears for (26).

Finally, good market clearing, which holds by Walras' law when other markets are clear, is defined as:

$$Y_t = C_t + G_t + I_t + C^E_t + \mu G(\bar{\omega}^F_t) R^K_t q_{t-1} K_t + \Upsilon_t \ , \tag{43}$$

where on the left-hand side we have total output. On the right-hand side, apart from household consumption,[20] public expenditure and investments, we also find entrepreneurial consumption, C^E (due to dying entrepreneurs), auditing costs for investment banks, and

[20] Recall that household goods consumption consists of composite consumption and leisure. For aggregate quantities: $C_t = X_t + \frac{L_t^{1+\gamma}}{1+\gamma}$

resources used for household loans, $\Upsilon_t = \Xi_t(B_t) + \omega_t^H B_t$. [21]

3 Numerical implementation

To solve the model, I follow the solution proposed in Bayer et al. (2019) and Luetticke (2021). As the joint distribution, Θ_t, is an infinite-dimensional object (and therefore not computable), it is discretized and represented by its histogram, a finite-dimensional object. I solve the households' policy function using the Endogenous Grid-point Method (EGM) developed by Carroll (2006), iterating over the first-order condition and approximating the idiosyncratic productivity process using a discrete Markov chain with four states using the Tauchen (1986) method. The log grid for liquid assets comprises of 100 points. I solve for aggregate dynamics by first-order perturbation around the steady-state, as in Reiter (2009).Differently from Bayer et al. (2019) and Luetticke (2021), I do not need to resort to the dimensionality reduction developed by Bayer and Luetticke (2020), because my joint distribution is represented by a bi-dimensional matrix (capital K does not display heterogeneity) with a total of 400 grid points, maintaining a sufficiently low computational time.

4 Calibration

The model is calibrated on the US economy, and because the focus is on conventional monetary policy, business cycle moments are targeted on the *Great Moderation* (i.e. 1985-2007). Periods in the model represent quarters; consequently, the following values for the calibrated parameters are intended quarterly, unless otherwise specified. Table 1 provides a list of calibrated parameters for the model, whereas Table 2 shows the model's effectiveness in replicating wealth distribution and business cycle moments.

4.1 Households

For the households' utility function, I assume the coefficient of relative risk aversion $\xi = 4$, as in Bayer et al. (2019). I set the Frisch elasticity of labor supply $\gamma = 1$, in line with the results of Chetty et al. (2011). The intertemporal discount factor, β, is equal to 0.988, so that deposits in investment banks are sufficient to have a leverage for entrepreneurs of 2, the same value used by Bernanke et al. (1999) in their model. The borrowing limit, $\underline{a}$, is set such that 16% of households have a negative wealth position, a value in line with empirical data from the Survey of Consumer Finances (SCF) 1983–2007 (Luetticke, 2021).

[21] Similarly to Kaplan et al. (2018), we can consider the last two terms in (43) as expenses for "financial services".

Table 1: Calibrated parameters

Parameter	Value	Description
β	0.988	Discount factor
ξ	4	Relative risk aversion
γ	1	Frisch elasticity of labor
$\underline{a}$	-2.6	Borrowing constraint
ι	0.0625	Prob. of leaving entr. state
ζ	0.00115	Prob. become rentier
ρ_h	0.98	Persistence of idio. prod. shock
σ_h	0.06	SD if idio. prod. shock
α	0.7	Labor share of production
δ	1.35%	Depreciation rate
η	20	Elasticity of substitution
κ	0.09	Price stickiness
ϕ	7.5	Adjustment cost of capital
γ^F	0.986	Entr. surviving rate
ρ_z	0.95	TFP shock persistence
σ_z	0.87%	TFP shock SD
R	1.005	Nominal int. rate
ρ^R	0.8	Int. rate smoothing
ρ^π	1.5	Reaction to inflation
σ_R	0.25% - 0.14%	Monetary shock SD
τ	0.3	tax rate
ρ_{gov}	0.86	Auto-correlation of debt
η^{FF}	51.62	convex technology for HHs loans
$\tilde{\Xi}$	$1.26e^{29}$	comm. bank loans parameter
μ	0.12	Auditing costs
σ_ω	0.27	SD of the id. shock on entr.

Table 2: Wealth distribution and business cycle moments

Wealth distribution moments

Target	Model	Target
Gini wealth	0.78	0.78
Share of borrowers	0.16	0.16
Top 10% wealth	0.69	0.67

Business cycle moments

Target	Model	Target
SD of Y (%)	1.38	1.38
σ^I/σ^Y	3	3
SD of C (%)	0.64	0.98
Corr. of Y with Y	1	1
Corr. of I with Y	0.99	0.92
Corr. of C with Y	0.99	0.92

Real GDP, investment and consumption are in logs. All data for business cycle moment analysis are processed with a H-P filter with $\lambda = 1600$. The calibrated moments for wealth distribution are the Gini index for wealth and the share of borrowing households. For business cycle moments, SD of Y and SD of I after a TFP shock.

The calibration of the productivity transition matrix, which determines how households move between the worker and rentier states, aims to provide a distribution of wealth consistent with empirical data. As in Luetticke (2021), I assume that the probability of becoming a rentier is the same for workers independent of their labor productivity, and that once they become workers again, they start with median productivity. The probability of leaving the rentier state is $\iota = 0.0625$, following the findings of Guvenen et al. (2014) on the probability of dropping out of the top 1% income group in the US. The probability of moving from the worker to the rentier state is set to $\zeta = 0.00115$, a value calibrated to obtain a Gini coefficient for wealth of 78% (in line with data from the SCF), which implies a share of rentier households equal to 1.8%. Regarding idiosyncratic income risk for labor productivity, I set autocorrelation $\rho_h = 0.98$ and standard deviation $\sigma_h = 0.06$, as estimated by Bayer et al. (2019).

4.2 Financial intermediaries

I target the two spreads for the financial frictions to be equal to 2% p.a.. The reasons for this choice are twofold. First, it involves comparison purposes between the two scenarios. Second, they are the same values used in both Bernanke et al. (1999) and Cúrdia and Woodford (2016), allowing the model to be consistent with the existing literature.

Regarding commercial banks, I follow Cúrdia and Woodford (2016), assuming that a one-percent increase in the volume of credit increases the borrowing spread by one percentage point p.a.. Together with the targeted value $\omega^H = 0.005$, this implies $\eta^{FF} = 51.62$ and $\tilde{\Xi} = 1.26e^{29}$.

The parameters concerning financial frictions on firms are in the ballpark of Bernanke et al. (1999) calibrations; therefore, the auditing cost is $\mu = 0.12$ and the standard deviation of the idiosyncratic shock on the entrepreneur's returns is $\sigma_\omega = 0.27$, which are calibrated to have $EFP_t = 1.005$ when the corporate leverage is 2. The share of surviving entrepreneurs, γ^F, is calibrated such that, at steady-state, the equity level in (22) is equal to the equity implied by (20).

4.3 Production Sector

The labor share of production (accounting for profits) and capital depreciation rate follow standard values in the literature and are set respectively to $\alpha = 0.7$ and $\delta = 1.35\%$. The mark-up is also standard, at 5%, which implies elasticity of substitution between goods varieties $\eta = 20$. The price stickiness parameter in the NKPC, $\kappa = 0.09$, is calibrated to generate a slope of the curve similar to the one that would arise in a model with sticky prices à la Calvo, with an average price duration of four quarters. The adjustment cost of capital parameter is calibrated to $\phi = 7.5$ to obtain investment-to-output volatility of 3 after a TFP shock, a standard value for U.S. data, in a scenario where none of the

frictions are active. The persistence of the TFP shock is $\rho_z = 0.95$, while the standard deviation is approximately $\sigma_z = 0.09$, standard measures in the literature.

4.4 Central Bank and Government

Inflation at the steady-state is set to 0, and the nominal (therefore real) interest rate on government bonds is 2%, a value in line with the real average federal funds rate for the Great Moderation period. I impose the same interest rate on all types of liquid savings (i.e., bonds and bank deposits); otherwise, households would choose to invest only in one asset or the other. Regarding the Taylor rule adopted by the Central Bank, the parameter for interest rate smoothing is $\rho_R = 0.8$, according to findings by Clarida et al. (2000), whereas the reaction to inflation fluctuations from the steady-state is $\rho_\pi = 1.5$, which is a common value in the macroeconomic literature.

For comparison purposes, I apply two different magnitudes for the monetary policy shock in the two scenarios. The standard deviation of the monetary policy shock for the case with financial frictions on household borrowing ability is $\sigma_R = 1\%$ p.a. (i.e., 0.25% quarterly). I then calibrate the shock for the other scenario to have a similar fluctuations in output between the two cases, delivering a parameter $\sigma_R = 0.14\%$ quarterly. The persistence of the shock is zero, implying that it is a one-time innovation.

The taxes set by the government are proportional to labor income and profits, with a tax rate $\tau = 0.3$ that targets the ratio of government spending to GDP to a standard value in the New Keynesian literature, approximately $G/Y = 20\%$. Since I am using a fiscal policy rule similar to the one adopted by Bayer et al. (2019), I also follow their estimation and set $\rho_B = 0.86$. This implies that most of the fiscal dynamic goes through government debt, with public spending adjusting to re-stabilize debt to its steady-state level.

5 Results

Before moving to inequality analyses, I examine aggregate fluctuations following the contractionary monetary policy shock. These results are not only useful for checking the consistency of my findings with the related literature, but also provide hints on differences at the idiosyncratic level between the two scenarios.

5.1 Aggregate fluctuations

As mentioned in Section 4.4, I assume two different magnitudes for monetary contraction. As a matter of fact, the two financial frictions are different in their construction, with the one affecting firms being more complex and delivering a higher level of financial acceleration. Therefore, I believe that a fairer comparison would be between two shocks that

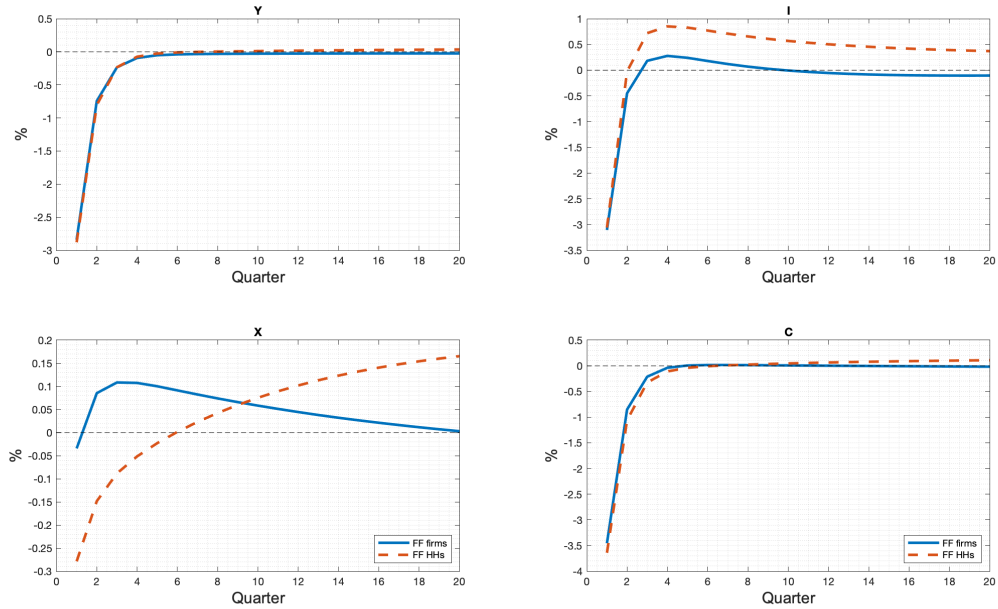

Figure 1: Impulse response to a monetary contraction for aggregate variables.

Note: monetary shock $\epsilon^R = 0.0025$ for active financial frictions on household borrowing, $\epsilon^R = 0.0014$ otherwise. The blue line refers to an economy with financial frictions on firms, the red one when frictions are on households.

have similar effects on the economy, rather than between two identical shocks. However, as Appendix D shows, considering the same magnitude for the monetary policy shock does not substantially change the underlying dynamics. Figure 1 shows the response for some aggregate variables.[22]

As calibrated, the output responses in the two cases are almost identical and negative. Aggregate investment fluctuations follow similar patterns as well. The changes in aggregate goods consumption, C, are also similar, but the same cannot be said for composite consumption, X. A possible explanation can be found in the impulse responses for labor, L, shown in Appendix C. Recalling that goods consumption comprises both composite consumption and leisure (which is directly proportional to labor), fluctuations in labor are much stronger than fluctuations in composite consumption. Because impulse responses for labor are substantially identical between the two cases, fluctuations in goods consumption end up having a shape more similar to those for labor, and therefore do not change much between the two scenarios. Thus, even in view of (10), I decide to focus on analyzing the dynamics of composite consumption, X, to better understand the differences between the two financial frictions.[23]

Composite consumption fluctuations show evident differences in terms of behavior.

[22]Since the analysis of aggregate fluctuation is not at the core of this research, I decided to show in the main text impulse responses only for four variables. More aggregate impulse responses can be found in Appendix C.

[23]In Appendix E, I provide inequality analyses for goods consumption, C. The results are in line with those of composite consumption.

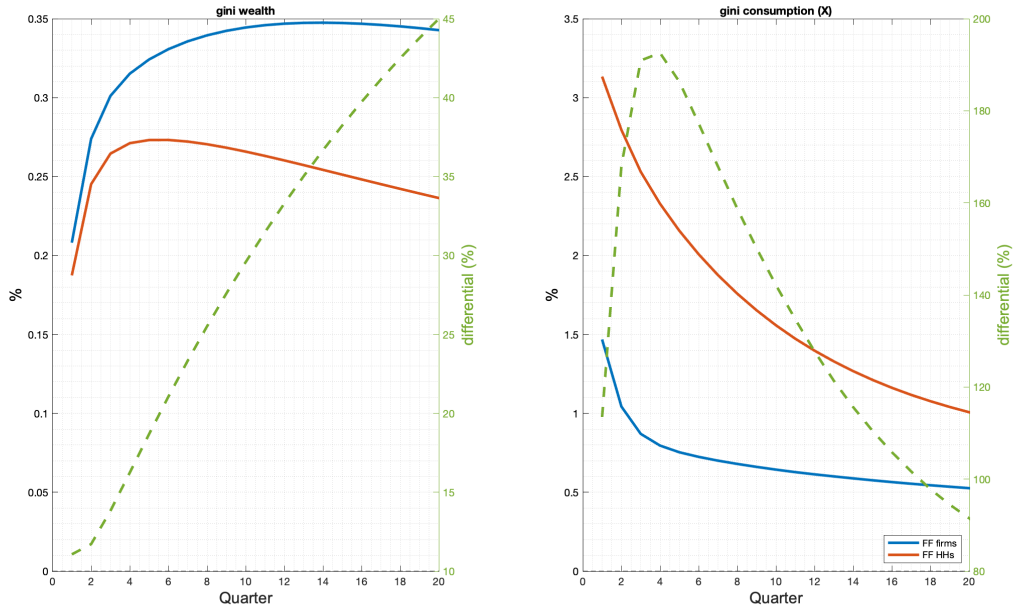

Figure 2: Impulse responses to a monetary contraction for wealth and consumption inequality.

Note: monetary shock $\epsilon^R = 0.0025$ for active financial frictions on household borrowing, $\epsilon^R = 0.0014$ otherwise. The blue line refers to an economy with financial frictions on firms, the red one when frictions are on households. The green dotted line is the percentage difference between the two curves using the lower curve as the base.

Even though the consumption response is negative in both cases on impact, the fall is much muted in the case of financial frictions on firms, and quickly overshoots. On the other hand, when considering financial frictions on households, composite consumption depression is more important and persistent, lasting more than a year before overshooting. This result suggests that even if different frictions have almost identical results in terms of output and investment, they could still affect household consumption choices differently. This mechanism is clearer in Section 5.4, where aggregate composite consumption is decomposed.

5.2 Wealth and consumption inequality

Using the Gini index of inequality for wealth and composite consumption, I now answer the initial question posed in this paper, that is, whether financial frictions affect the household distribution of wealth and consumption differently after a contractionary monetary policy shock. In Figure 2, the evolution of the two indices for the two cases (blue and red solid lines) and the green dotted line representing the difference between the two curves in percentage terms are shown.

The first consideration that can be done is that there is a significant differences between the two curves for both wealth and consumption. In the case of wealth, this difference is lower on impact, around 10%, but steadily increases over time. The pattern

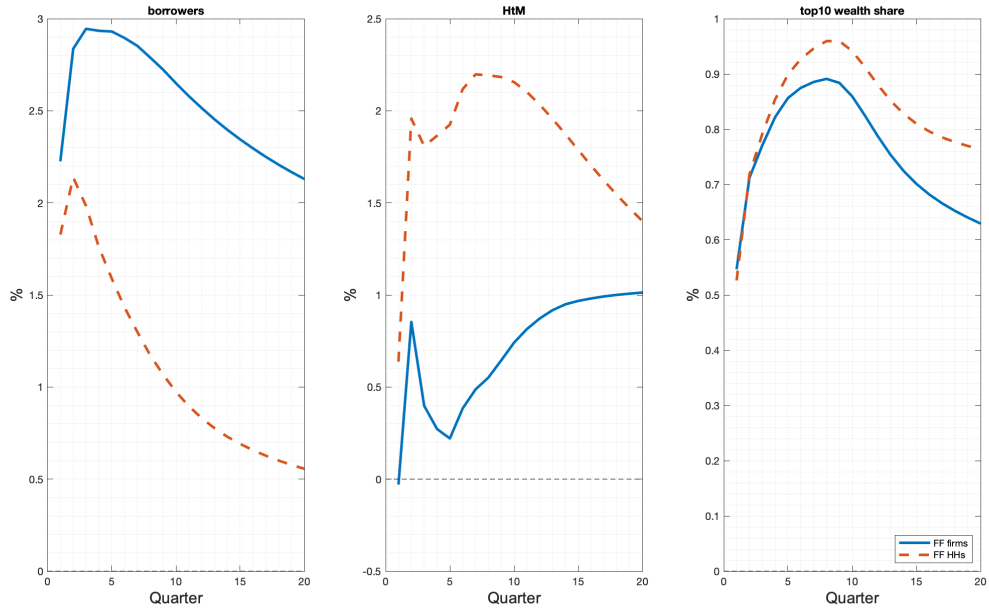

Figure 3: IRFs for the share of borrowing households, HtM households and wealth held by the top 10%

Note: monetary shock $\epsilon^R = 0.0025$ for active financial frictions on household borrowing, $\epsilon^R = 0.0014$ otherwise. The blue line refers to an economy with financial frictions on firms, the red one when frictions are on households.

of the consumption difference is hump-shaped, but higher in magnitude, with a peak of approximately 190%. The second, and probably the most important, consideration is that while financial frictions on firms implies a higher degree of wealth inequality, the same does not happen in the case of consumption inequality, with the Gini index for the case of friction on household borrowing being significantly greater. This means that although wealth and consumption inequalities increase in both cases, they do so at a relatively inverted pace. The Gini index does not indicate how wealth and consumption are scattered among the agents. Therefore, to understand why this happens, I first analyze which share of households is more responsible for the difference in wealth results between the two cases.

5.3 Wealth dynamics

To decompose wealth inequality, I compute impulse responses for three measures: share of borrowing households, share of Hand-to-Mouth (HtM) households,[24] and share of wealth held by the richest 10% of the distribution. The results are shown in Figure 3.

For what concerns the top tail of the distribution, wealth in the hands of the top 10%

[24]In theoretical literature, HtM households are usually considered agents with zero wealth. However, in my model, a relevant number of grid points for wealth is extremely close to the zero-wealth threshold. Therefore, I choose to consider as HtM those households with zero wealth or positive wealth that do not exceed the quarterly lower possible realization of labor income.

increases similarly in the two cases, at least for the first year, and then sees slightly higher values when financial frictions are active on household borrowings.[25] The bottom of the distribution, however, shows interesting results that could explain why we have higher inequality in the case of frictions in the production sector. In this scenario, the share of borrowers increases relatively more. However, the opposite occurs for the share of HtM households.

Since the richer top 10% holds most of the wealth (69% in my model) and the dynamics for this share of households are quite similar, it is plausible that the difference in the wealth Gini index fluctuations is mostly due to what happens between borrowing and HtM households. The negative shock worsens labor conditions, especially affecting poorer households, since they rely mainly on labor income for consumption and saving. In response to the worsening of their income, households would like to become borrowers to improve their consumption smoothing. However, in the case of financial frictions on household borrowing ability, a larger aggregate household debt implies a higher borrowing spread. This means that, in the case of financial frictions on firms, borrowing is relatively cheaper; thus, households are encouraged to move to the bottom of the wealth distribution. On the contrary, when the household borrowing spread increases, households are less eager to become borrowers, and therefore have a relatively higher level of wealth. These movements in the wealth distribution are consistent with the results on the left-hand side of Figure 2. To determine whether this dynamic is plausible, I investigate the decomposition of aggregate composite consumption.

5.4 Consumption dynamics

To see how different shares of households respond to a rise in the nominal interest rate in terms of consumption, I decompose the impulse response of aggregate composite consumption into average consumption responses for specific shares of the population according to their wealth.[26] The results are shown in Figure 4, where, on the left-hand side, there is consumption decomposition for the case of financial frictions of firms, and the counterfactual scenario of friction on household borrowing on the right-hand side.

Households at the top the wealth distribution consume relatively more in case of financial frictions on households, but there are no striking differences in terms of impulse responses behavior. This again suggests that most of the action takes place at the lower end. A greater fall in average consumption can be observed in the case of financial frictions on households. For instance, on impact, the fall in average consumption for borrowers (red

[25] This latter effect could be due to the higher increase in the risk-free rate for the case of financial frictions on households, since all wealth in the model is liquid.

[26] I choose to use average consumption fluctuations instead of absolute consumption fluctuations because I think they are better suited for comparisons between the two cases, but also for comparisons within the same case with respect to aggregate consumption fluctuations. Note that aggregate consumption is the case limit where the average consumption for the whole population is considered.

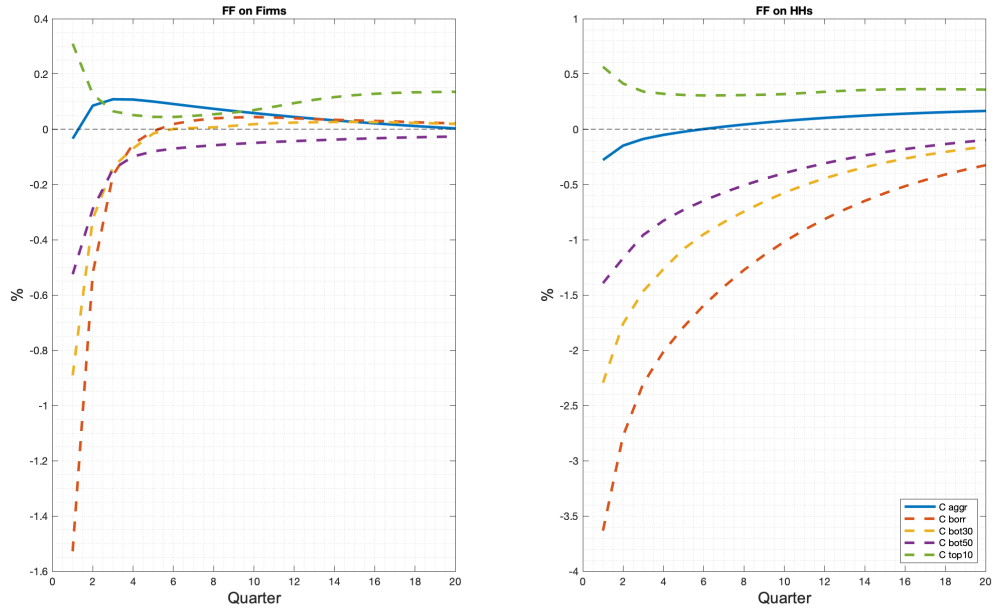

Figure 4: Average consumption fluctuation for different shares of households.
Note: monetary shock $\epsilon^R = 0.0025$ for active financial frictions on household borrowing, $\epsilon^R = 0.0014$ otherwise.

dotted line) is three times larger when compared to the case of frictions on firms. It can be inferred that poorer households actually consume less when the loan rate for household borrowing increases, which is in line with the findings in the previous section. A larger number of them decide to become HtM with very little or no possibility of consumption smoothing, and hence significantly decrease their consumption. This implies a higher consumption dispersion, explaining the greater Gini index for consumption obtained in the right-hand graph of Figure 2. On the contrary, in the case of financial frictions on firms, borrowers not only increase in number (as shown in Figure 3), but their average consumption overshoots after one year, suggesting that the borrowing penalty ω^H plays a crucial role in the behavior of these households.

To verify whether the increasing loan rate is indeed the main reason for this behavior or whether there are other forces in place, I follow Luetticke (2021) and decompose the monetary transmission by writing aggregate composite consumption as a function of a sequence of household policy functions induced by equilibrium prices relevant to household consumption choice, $\{\Omega_t\}_{t\geq 0}$, with $\Omega_t = \left\{\frac{R_t^I}{\pi_t}, W_t, \Pi_t\right\}$:

$$X_t\left(\{\Omega_t\}_{t\geq 0}\right) = \int x_t\left(a, h; \{\Omega_t\}_{t\geq 0}\right) d\Theta_t, \tag{44}$$

where $\Theta_t\left(da, dh; \{\Omega_t\}_{t\geq 0}\right)$ is the joint distribution of liquid assets and idiosyncratic labor productivity. Totally differentiating (44) decomposes the total response to monetary shocks into parts explained by each single price.[27] Result are shown in Figure 5.

[27] A similar decomposition can be found in Kaplan et al. (2018) and Auclert (2019).

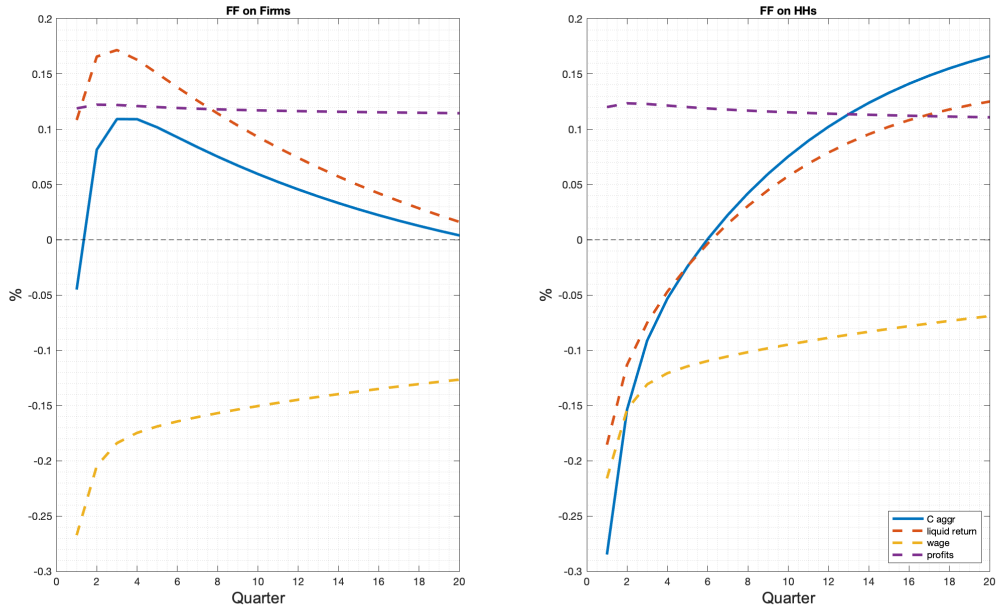

Figure 5: Consumption decomposition for relevant prices

Note: monetary shock $\epsilon^R = 0.0025$ for active financial frictions on household borrowing, $\epsilon^R = 0.0014$ otherwise. The graph on the left-hand side represents the decomposition for the case of friction on firms, while the one on the right-hand side represents the case of friction on household borrowing.

The profit contribution to the shape of consumption is almost identical. The wage contribution also has a similar pattern, although it is slightly stronger and more persistent in the case of financial frictions on firms. This is probably due to the friction affecting the production sector, which is the main source of income for most of the population, especially the poorer ones. The greatest difference between the two graphs is in the shape and magnitude of the contribution of returns on liquidity. In case of financial frictions on firms, it is positive and hump-shaped. One possible explanation is that the wealth owned by households in this model is exclusively liquid by construction. Therefore, while poorer households are slightly affected by this, richer households benefit enormously from an increase in the liquid return. In the case of financial friction on households, the contribution of liquidity returns is significantly negative for more than one year and its shape and magnitude are similar to those of aggregate consumption.

It must be noted that in Figure 5, the liquidity return includes the household borrowing spread, $(1 + \omega_t^H)$. Since this spread is changing in the case of financial frictions on households, to determine its significance I further decompose the liquidity return and consider the borrowing penalty ω^H as an individual price; the results are shown in Figure 6.

The household borrowing penalty plays a significant role in the consumption depression, which is even more significant than the wage contribution for at least a couple of years. The liquidity return net of the borrowing penalty has effects similar to those in

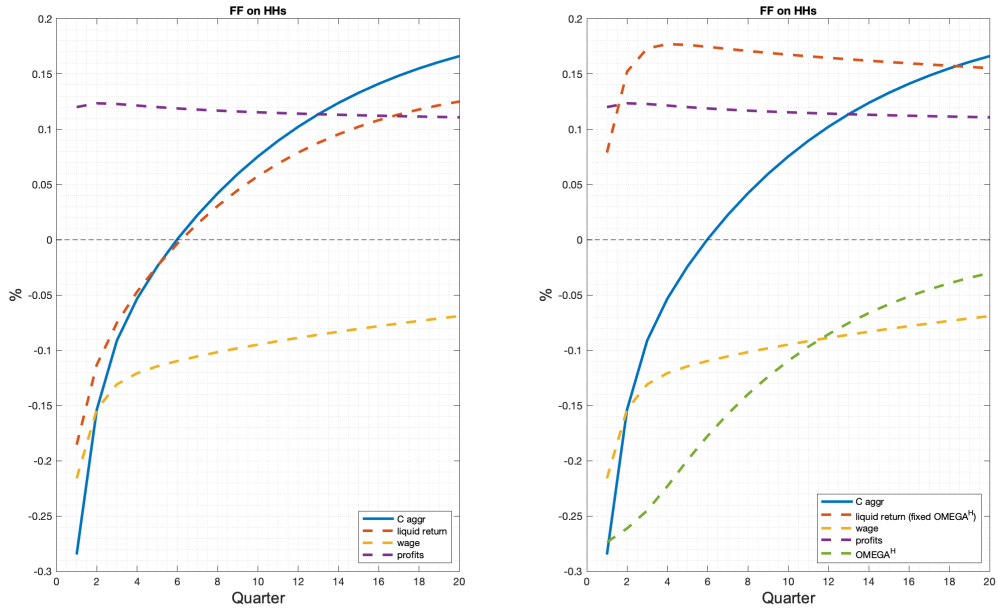

Figure 6: Consumption decomposition for relevant prices

Note: monetary shock $\epsilon^R = 0.0025$. The graph on the left-hand side represents the decomposition considering the household borrowing spread in the liquidity return. In the graph on the right-hand side, I consider the same case with the borrowing penalty (ω_t^H) as an individual variable.

the case of active frictions on firms. This finding validates the assumption that most of the differences in wealth and consumption inequality between the two cases can be explained by household behavior around the zero-wealth thresholds. More specifically, although firms' financial frictions have significant effects on household consumption in terms of labor income, most of the divergence between the two scenarios is due to the varying household borrowing spread and, therefore, financial frictions on households.

6 Concluding remarks

Building a full-fledged HANK model featuring two financial frictions that directly affect different players in the economy, I show that these frictions have different effects on households in terms of wealth and consumption dynamics. In the face of a contractionary monetary policy shock, wealth and consumption inequality increase in both cases; however, the pace at which this occurs is reversed. When financial frictions on firms are in place, the Gini index for wealth is relatively higher. By contrast, a higher dispersion in consumption occurs in the case of active friction on household borrowing.

The explanation for this phenomenon can be traced back to how households behave in the two scenarios around the zero-wealth threshold; that is, when they have to decide whether to be borrowers or Hand-to-Mouth. Active frictions on the productive side of the economy cause a decrease in labor income, which pushes poorer households further

down their wealth distribution because they mostly rely on labor for consumption and saving purposes. Because frictions on household borrowing are inactive in this case, it is relatively cheaper for them to be borrowers. Then, they can move in a larger quantity to the bottom of the distribution, generating a higher increase in wealth inequality but a lower one for consumption, since they can smooth consumption through borrowed liquidity. On the other hand, when the household borrowing spread increases after a monetary shock, borrowings are relatively more expensive for households. Therefore, a higher proportion of households choose to become HtM and consume their current income. This prevents them from going further down the distribution but also affects their consumption smoothing, resulting in higher consumption inequality.

Despite using a relatively simple model, this study provides interesting guidance in terms of the redistribution effects of monetary policy according to a possible state of the economy. Even though wealth redistribution should not be a formal target for central banks, policymakers have undoubtedly been concerned with it over the last years, and recent findings have proven that this concern is also important from a macro-modeling point of view.

Nonetheless, there may be several dimensions across which this study could be extended to future research. For instance, this is a model concerning conventional monetary policy, but after the Great Financial Crisis, interest rates hit the zero lower bound for a prolonged period, forcing central banks to resort to unconventional policies. Another important topic in current theoretical macroeconomics is the heterogeneity of productive firms, which could be an extremely interesting extension of this model, that is, how productive sector heterogeneity is important when certain frictions are present. In addition, I assume that the friction generated by commercial banks is exclusively due to the resources they waste for their operations, with the intent of keeping the model as simple as possible. However, Cúrdia and Woodford (2016) also assume that a certain quantity of household debt is subject to default. According to Nakajima and Ríos-Rull (2019), unsecured household credit (i.e., credit-bearing default risk) is necessary to reproduce procyclicality in household debt, as seen in the empirical data.

References

Aiyagari, S. R. (1994). Uninsured idiosyncratic risk and aggregate saving. *The Quarterly Journal of Economics*, 109(3):659–684.

Akerlof, G. A. (1970). The market for "lemons": Quality uncertainty and the market mechanism. *The Quarterly Journal of Economics*, 84(3):488–500.

Attanasio, O. P. and Weber, G. (1995). Is Consumption Growth Consistent with Intertemporal Optimization? Evidence from the Consumer Expenditure Survey. *Journal of Political Economy*, 103(6):1121–1157.

Auclert, A. (2019). Monetary policy and the redistribution channel. *American Economic Review*, 109(6):2333–67.

Auclert, A., Bardóczy, B., and Rognlie, M. (2023). MPCs, MPEs, and Multipliers: A Trilemma for New Keynesian Models. *The Review of Economics and Statistics*, 105(3):700–712.

Auclert, A., Bardóczy, B., Rognlie, M., and Straub, L. (2021). Using the sequence-space jacobian to solve and estimate heterogeneous-agent models. *Econometrica*, 89(5):2375–2408.

Bayer, C. and Luetticke, R. (2020). Solving heterogeneous agent models in discrete time with many idiosyncratic states by perturbation methods. *Quantitative Economics*, 11:1253–1288.

Bayer, C., Luetticke, R., Pham-Dao, L., and Tjaden, V. (2019). Precautionary savings, illiquid assets, and the aggregate consequences of shocks to household income risk. *Econometrica*, 87(1):255–290.

Bernanke, B. S., Gertler, M., and Gilchrist, S. (1999). The financial accelerator in a quantitative business cycle framework. In Taylor, J. B. and Woodford, M., editors, *Handbook of Macroeconomics*, volume 1 of *Handbook of Macroeconomics*, chapter 21, pages 1341–1393. Elsevier.

Brunnermeier, M. K., Eisenbach, T. M., and Sannikov, Y. (2012). Macroeconomics with financial frictions: A survey. Working Paper 18102, National Bureau of Economic Research.

Carlstrom, C. T., Fuerst, T. S., and Paustian, M. (2016). Optimal contracts, aggregate risk, and the financial accelerator. *American Economic Journal: Macroeconomics*, 8(1):119–47.

Carroll, C. D. (2006). The method of endogenous gridpoints for solving dynamic stochastic optimization problems. *Economics Letters*, 91(3):312–320.

Chatterjee, S., Corbae, D., Nakajima, M., and Rios-Rull, J.-V. (2007). A quantitative theory of unsecured consumer credit with risk of default. *Econometrica*, 75(6):1525–1589.

Chetty, N., Weber, A., Guren, A. M., and Day, M. (2011). Are Micro and Macro Labor Supply Elasticities Consistent? A Review of Evidence on the Intensive and Extensive Margins. Technical report.

Christiano, L. J., Motto, R., and Rostagno, M. (2014). Risk Shocks. *American Economic Review*, 104(1):27–65.

Clarida, R., Galí, J., and Gertler, M. (2000). Monetary policy rules and macroeconomic stability: Evidence and some theory. *The Quarterly Journal of Economics*, 115(1):147–180.

Coibion, O., Gorodnichenko, Y., Kueng, L., and Silvia, J. (2017). Innocent bystanders? monetary policy and inequality. *Journal of Monetary Economics*, 88:70–89.

Colciago, A., Samarina, A., and de Haan, J. (2019). Central bank policies and income and wealth inequality: A survey. *Journal of Economic Surveys*, 33(4):1199–1231.

Cúrdia, V. and Woodford, M. (2016). Credit Frictions and Optimal Monetary Policy. *Journal of Monetary Economics*, 84(C):30–65.

Farhi, E. and Tirole, J. (2012). Collective moral hazard, maturity mismatch, and systemic bailouts. *American Economic Review*, 102(1):60–93.

Ferlaino, F. (2023). Does the financial accelerator accelerate inequalities? *Cefes paper series*.

Fernández-Villaverde, J., Hurtado, S., and Nuño, G. (2023). Financial frictions and the wealth distribution. *Econometrica*, 91(3):869–901.

Fisher, I. (1933). The debt-deflation theory of great depressions. *Econometrica*, 1(4):337–357.

Gertler, M. and Karadi, P. (2011). A model of unconventional monetary policy. *Journal of Monetary Economics*, 58(1):17–34. Carnegie-Rochester Conference Series on Public Policy: The Future of Central Banking April 16-17, 2010.

Gilchrist, S. and Zakrajšek, E. (2012). Credit spreads and business cycle fluctuations. *American Economic Review*, 102(4):1692–1720.

Greenwood, J., Hercowitz, Z., and Huffman, G. (1988). Investment, capacity utilization, and the real business cycle. *American Economic Review*, 78(3):402–17.

Guerrieri, V. and Lorenzoni, G. (2017). Credit crises, precautionary savings, and the liquidity trap. *The Quarterly Journal of Economics*, 132(3):1427–1467.

Guvenen, F., Kaplan, G., and Song, J. (2014). The glass ceiling and the paper floor: Gender differences among top earners, 1981-2012. Working Paper 20560, National Bureau of Economic Research.

Holmstrom, B. and Tirole, J. (1997). Financial intermediation, loanable funds, and the real sector. *The Quarterly Journal of Economics*, 112(3):663–691.

Huggett, M. (1993). The risk-free rate in heterogeneous-agent incomplete-insurance economies. *Journal of Economic Dynamics and Control*, 17(5):953–969.

Iacoviello, M. (2005). House prices, borrowing constraints, and monetary policy in the business cycle. *American Economic Review*, 95(3):739–764.

Imrohoroğlu, A. (1989). Cost of business cycles with indivisibilities and liquidity constraints. *Journal of Political Economy*, 97(6):1364–1383.

Jordà (2005). Estimation and inference of impulse responses by local projections. *The American Economic Review*, 95(1):161–182.

Kaplan, G., Moll, B., and Violante, G. L. (2018). Monetary policy according to hank. *American Economic Review*, 108(3):697–743.

Krusell, P. and Smith, Jr., A. (1998). Income and wealth heterogeneity in the macroeconomy. *Journal of Political Economy*, 106(5):867–896.

Lee, S., Luetticke, R., and Ravn, M. (2020). Financial frictions: Macro vs micro volatility. CEPR Discussion Papers 15133.

Luetticke, R. (2021). Transmission of monetary policy with heterogeneity in household portfolios. *American Economic Journal: Macroeconomics*, 13(2):1–25.

Nakajima, M. and Ríos-Rull, J.-V. (2019). Credit, Bankruptcy, and Aggregate Fluctuations. Working Papers 19-48, Federal Reserve Bank of Philadelphia.

Ottonello, P. and Winberry, T. (2020). Financial heterogeneity and the investment channel of monetary policy. *Econometrica*, 88(6):2473–2502.

Piketty, T. (2017). *Capital in the Twenty-First Century*. Harvard University Press.

Reiter, M. (2009). Solving heterogeneous-agent models by projection and perturbation. *Journal of Economic Dynamics and Control*, 33(3):649–665.

Rotemberg, J. J. (1982). Sticky prices in the united states. *Journal of Political Economy*, 90(6):1187–1211.

Stiglitz, J. E. and Weiss, A. (1981). Credit rationing in markets with imperfect information. *The American Economic Review*, 71(3):393–410.

Storesletten, K., Telmer, C., and Yaron, A. (2004). Cyclical dynamics in idiosyncratic labor market risk. *Journal of Political Economy*, 112(3):695–717.

Tauchen, G. (1986). Finite state markov-chain approximations to univariate and vector autoregressions. *Economics Letters*, 20(2):177–181.

Townsend, R. M. (1979). Optimal contracts and competitive markets with costly state verification. *Journal of Economic Theory*, 21(2):265–293.

Appendix

A Idiosyncratic productivity process and the joint distribution

Households can be workers, with productivity $h > 0$, or rentiers, , with $h = 0$, which means that they do not earn labor income but only profit income. Furthermore, I assume that there are three possible productivity realizations for workers: high productivity, h^H, median productivity, h^M, and low productivity, h^L. The Markov process generates the following transition matrix:

$$
\begin{array}{cc}
 & h_{t+1} \\
 & \begin{array}{cccc} h^L & h^M & h^H & 0 \end{array} \\
h_t \begin{array}{c} h^L \\ h^M \\ h^H \\ 0 \end{array} & \begin{bmatrix} p_{LL}(1-\zeta) & p_{LM}(1-\zeta) & p_{LH}(1-\zeta) & \zeta \\ p_{ML}(1-\zeta) & p_{MM}(1-\zeta) & p_{MH}(1-\zeta) & \zeta \\ p_{HL}(1-\zeta) & p_{HM}(1-\zeta) & p_{HH}(1-\zeta) & \zeta \\ 0 & \iota & 0 & 1-\iota \end{bmatrix}
\end{array}
$$

with probabilities, p, determined using the Tauchen method. I follow other studies using this household distribution framework, such as Bayer et al. (2019) and Luetticke (2021), and assume that rentiers who become workers are endowed with the median productivity level ($h = 1$).

At the steady state, a joint distribution of households exists according to their wealth level, a, and their productivity, h. This joint distribution can be represented by the bi-dimensional matrix as follows:

$$
\text{prod. } h \begin{array}{c} h_m \\ \cdots \\ h_2 \\ h_1 \end{array} \begin{bmatrix} H_{m,1} & H_{m,2} & \cdots & H_{m,n} \\ \cdots & \cdots & \cdots & \cdots \\ H_{2,1} & H_{2,2} & \cdots & H_{2,n} \\ H_{1,1} & H_{1,2} & \cdots & H_{1,n} \end{bmatrix}
$$

$$
\begin{array}{cccc} a_1 & a_2 & \cdots & a_n \end{array}
$$

wealth a

where $H_{1,1}$ is the share of households with the lowest level of wealth and labor productivity (except for the last state $h_m = 0$, since in this model they are rentiers), and $\int H da dh = 1$. As the vector indicating possible household wealth levels is composed of 100 entries, this joint distribution matrix comprises 400 grid points ($a_n = 100$ and $h_m = 4$).

B Investment banks optimal contract

B.1 Idiosyncratic shock on return on capital

Following Bernanke et al. (1999), I assume that the Idiosyncratic shock ω^F is distributed log-normally. i.e. $\omega^F \in [0, +\infty)$.[28] Using results from Appendix A.2 in Bernanke et al. (1999) I can write $F(\omega^F)$, $\Gamma(\omega^F)$ and $G(\omega^F)$ in the analytical expressions that I use in my code to solve the model:

$$F(\omega^F) = \Phi\left[\left(log(\bar{\omega^F}) + \frac{1}{2}\sigma^2_{\omega^F}\right)/\sigma_{\omega^F}\right] , \tag{A1}$$

$$\Gamma(\omega^F) = \Phi\left[\left(log(\bar{\omega}^F) - \frac{1}{2}\sigma^2_{\omega^F}\right)/\sigma_{\omega^F}\right] + \bar{\omega}^F\left\{1 - \Phi\left[\left(log(\bar{\omega}^F) + \frac{1}{2}\sigma^2_{\omega^F}\right)/\sigma_{\omega^F}\right]\right\} , \tag{A2}$$

$$G(\omega^F) = \Phi\left[\left(log(\bar{\omega}^F) + \frac{1}{2}\sigma^2_{\omega^F}\right)/\sigma_{\omega^F} - \sigma_{\omega^F}\right] , \tag{A3}$$

with $\Phi(\cdot)$ being the normal cumulative distribution function and σ_{ω^F} the standard deviation of the idiosyncratic shock on entrepreneurs' return on capital.

B.2 Investment banks' participation constraint and entrepreneur j's optimization problem

After substituting (18) and (17) into (19), I obtain:

$$[1-F(\bar{\omega}^F_{jt+1})]\bar{\omega}^F_{jt+1}R^K_{t+1}q_tK_{jt+1}+(1-\mu)\int_0^{\bar{\omega}^F_{jt+1}}\omega^F_j dF(\omega^F_j)R^K_{t+1}q_tK_{jt+1} = \frac{R_{t+1}}{\pi_{t+1}}(q_tK_{jt+1}-N_{jt+1}) . \tag{A4}$$

Divide everything by $R^R_{t+1}q_tK_{jt+1}$:

$$\frac{R^K_{t+1}}{\frac{R_{t+1}}{\pi_{t+1}}}\left([1 - F(\bar{\omega}^F_{jt+1})]\bar{\omega}^F_{jt+1} + (1-\mu)\int_0^{\bar{\omega}^F_{jt+1}}\omega^F_j dF(\omega^F_j)\right) = \left(1 - \frac{N_{jt+1}}{q_tK_{jt+1}}\right) . \tag{A5}$$

Following the notation used in Bernanke et al. (1999) and Christiano et al. (2014):

[28] Note that other kinds of distribution with values greater or equal to 0 could be used as well. Here I choose to adapt the same distribution to give a sense of continuity between the two studies.

$$\Gamma(\bar{\omega}_j^F) \equiv \int_0^{\bar{\omega}_j^F} \omega_j^F dF(\omega_j^F) + \bar{\omega}_j^F \int_{\bar{\omega}_j^F}^{\infty} dF(\omega_j^F) \ , \qquad \mu G(\bar{\omega}_j^F) \equiv \mu \int_0^{\bar{\omega}_j^F} \omega_j^F dF(\omega_j^F) \ , \quad \text{(A6)}$$

where $\Gamma(\bar{\omega}_j^F)$ is the expected gross share of profits going to the lender and $\mu G(\bar{\omega}_j^F)$ is the expected monitoring cost paid by the lender. $\Gamma(\bar{\omega}_j^F)$ can be rewritten as:

$$\Gamma(\bar{\omega}_j^F) = G(\bar{\omega}_j^F) + \bar{\omega}_j^F \left[1 - F(\bar{\omega}_j^F)\right] \ . \quad \text{(A7)}$$

I can now use (A6) and (A7) in (A5) and rearrange to finally obtain:

$$\frac{R_{t+1}^K}{\left(\frac{R_{t+1}}{\pi_{t+1}}\right)} = \frac{1}{\Gamma(\bar{\omega}_{jt+1}^F) - \mu G(\bar{\omega}_{jt+1}^F)} \left(1 - \frac{N_{jt+1}}{q_t K_{jt+1}}\right) \ , \quad \text{(A8)}$$

where $\Gamma(\bar{\omega}_{jt+1}^F) - \mu G(\bar{\omega}_{jt+1}^F)$ is the share of entrepreneur j's profits going to the lender (as loan repayment), net of auditing costs.

Equation (A8) is the complete version of (20), which explain the function underlying $f\left(\bar{\omega}_{jt+1}^F, LEV_{jt+1}\right)$. For a higher level of entrepreneur leverage, the EFP increases, raising the return on capital. However, it also increases the probability of an entrepreneur's default, thereby increasing the net share of profit demanded by investment banks as loan repayment, resulting in higher financing costs for entrepreneurs. To see in detail how this mechanism works, I show the entrepreneur j's optimization problem below.

According to the optimal contract set by investment banks, entrepreneur j's expected return can be expressed as:

$$E_t \left\{ \int_{\bar{\omega}_{jt+1}^F}^{\infty} \omega_j^F dF(\omega_j^F) R_{t+1}^K q_t K_{jt+1} - (1 - F(\bar{\omega}_j^F)) R_{t+1}^K q_t K_{jt+1} \right\} \ , \quad \text{(A9)}$$

with expectations taken with respect to the realization of R_{t+1}^K. The first term of (A9) represents the entrepreneur's profit when she does not default on debt, while the second term is the amount of profits that she uses to repay the lender. Following the notation used above, and considering that the entrepreneur's return is subject to the participation constraint (19), I write entrepreneur j's optimal contracting problem as:

$$\begin{aligned} &\max_{\{K_{jt+1}, \bar{\omega}_{jt+1}^F\}} E_t \left\{ \left[1 - \Gamma(\bar{\omega}_{jt+1}^F)\right] R_{t+1}^K q_t K_{jt+1} \right\} \ , \\ &s.t. \quad \frac{R_{t+1}}{\pi_{t+1}} (q_t K_{jt+1} - N_{jt+1}) = \left[\Gamma(\bar{\omega}_{jt+1}^F) - \mu G(\bar{\omega}_{jt+1}^F)\right] R_{t+1}^K q_t K_{jt+1} \ . \end{aligned} \quad \text{(A10)}$$

Deriving F.O.C. I obtain:

$$w.r.t.\ \ \omega^F_{jt+1}: \quad -\Gamma'(\bar{\omega}^F_{jt+1}) + \lambda_{jt+1}\left[\Gamma'(\bar{\omega}^F_{jt+1}) - \mu G'(\bar{\omega}^F_{jt+1})\right] = 0\ , \tag{A11}$$

$$w.r.t.\ \ K_{jt+1}: \quad E_t\left\{\left[1-\Gamma(\bar{\omega}^F_{jt+1})\right]R^K_{t+1} - \lambda_{jt+1}\left[\frac{R_{t+1}}{\pi_{t+1}} - \left(\Gamma(\bar{\omega}^F_{jt+1}) - \mu G(\bar{\omega}^F_{jt+1})R^K_{t+1}\right)\right]\right\} = 0\ , \tag{A12}$$

$$w.r.t.\ \ \lambda_{jt+1}: \quad E_t\left\{\frac{R_{t+1}}{\pi_{t+1}}(q_tK_{jt+1} - N_{jt+1}) - \left[\Gamma(\bar{\omega}^F_{jt+1}) - \mu G(\bar{\omega}^F_{jt+1})\right]R^K_{t+1}q_tK_{jt+1}\right\} = 0\ , \tag{A13}$$

where λ_j is the Lagrangian multiplier for entrepreneur j's problem. By rearranging (A11), it is possible to express λ_{jt+1} as a function of only $\bar{\omega}^F_{jt+1}$. Furthermore, rearranging (A12):

$$E_t\left\{\frac{R^K_{t+1}}{\frac{R_{t+1}}{\pi_{t+1}}}\right\} = \frac{\lambda_{jt+1}}{\left[1-\Gamma(\bar{\omega}^F_{jt+1}) + \lambda_{jt+1}\left(\Gamma(\bar{\omega}^F_{jt+1}) - \mu G(\bar{\omega}^F_{jt+1})\right)\right]}\ . \tag{A14}$$

It can be proven that there is a monotonically increasing relationship between the EFP and $\bar{\omega}^F_j$. According to (A8), we can extend this relationship between the EFP and the leverage level of j, assessing that a higher entrepreneur's leverage implies a higher EFP.[29]

Furthermore, it is clear from (A14) that $\bar{\omega}^F_j$ is determined only by aggregate variables. Thus, any entrepreneur chooses the same threshold $\bar{\omega}^F$ for the idiosyncratic shock on capital returns, below which they default, and the same leverage level.[30] This result allows to consider only the aggregate variables in the production sector part of the model, since every entrepreneur has the same firm structure.

C Impulse responses of MP contractionary shock

Figure C.1 show several aggregate variables impulse responses for the monetary policy shock considered in the baseline model. This integrate Impulse Response Functions (IRFs) present in Figure 1 in the main text.

[29] See Appendix A.1 in Bernanke et al. (1999) for proofs.

[30] According to (A8), leverage is a function of the EFP (composed of only aggregate variables) and $\bar{\omega}^F_j$. If $\bar{\omega}^F_j$ depends only on aggregate variables (since it is a function of the EFP, according to (A14)), then the same can be said for the leverage.

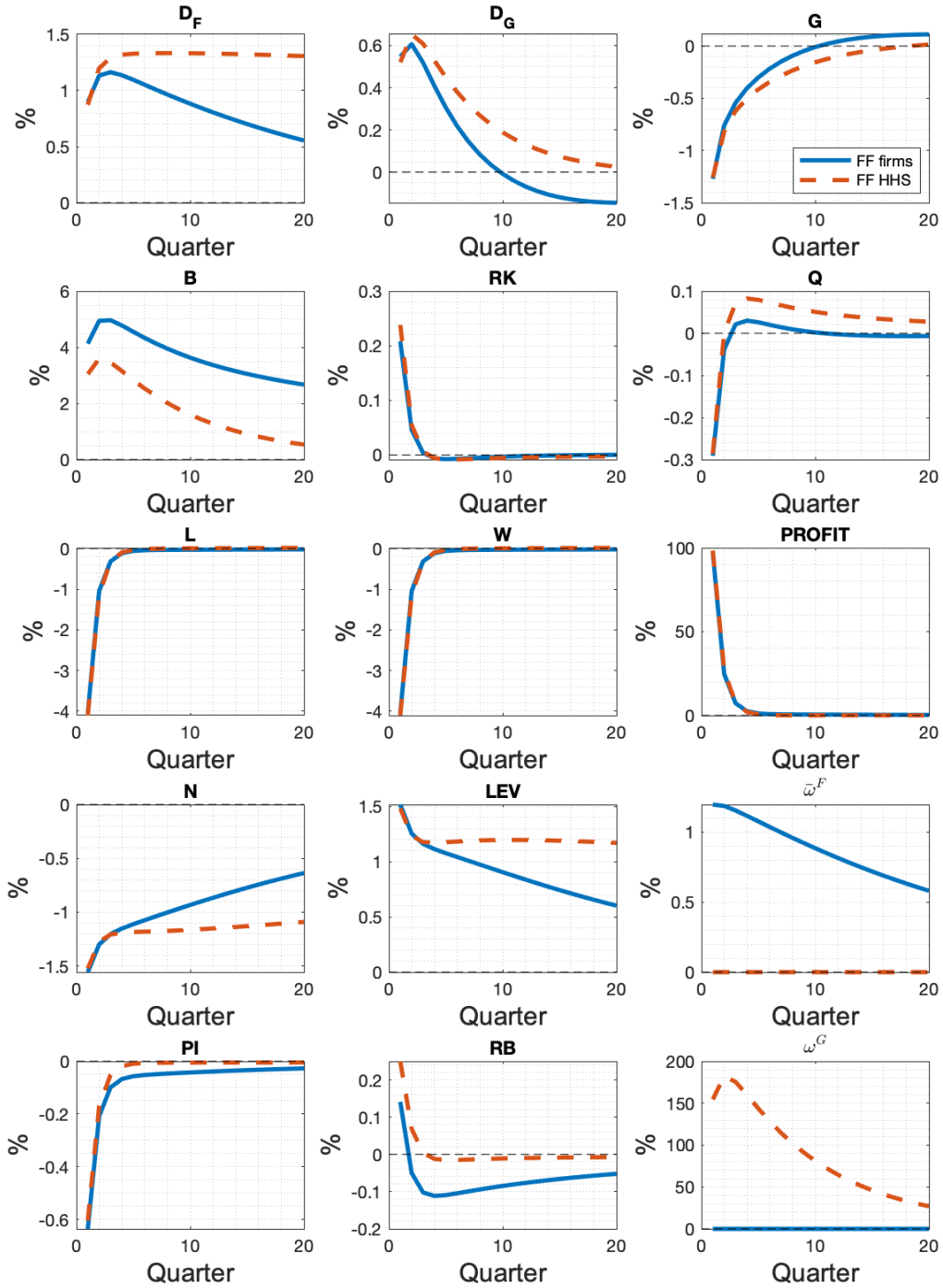

Figure C.1: Impulse response to a monetary contraction for aggregate variables

Note: monetary shock $\epsilon^R = 0.0025$ for active financial frictions on household borrowing, $\epsilon^R = 0.0014$ otherwise. The blue line refers to an economy with financial frictions on firms, the red one when frictions are on households.

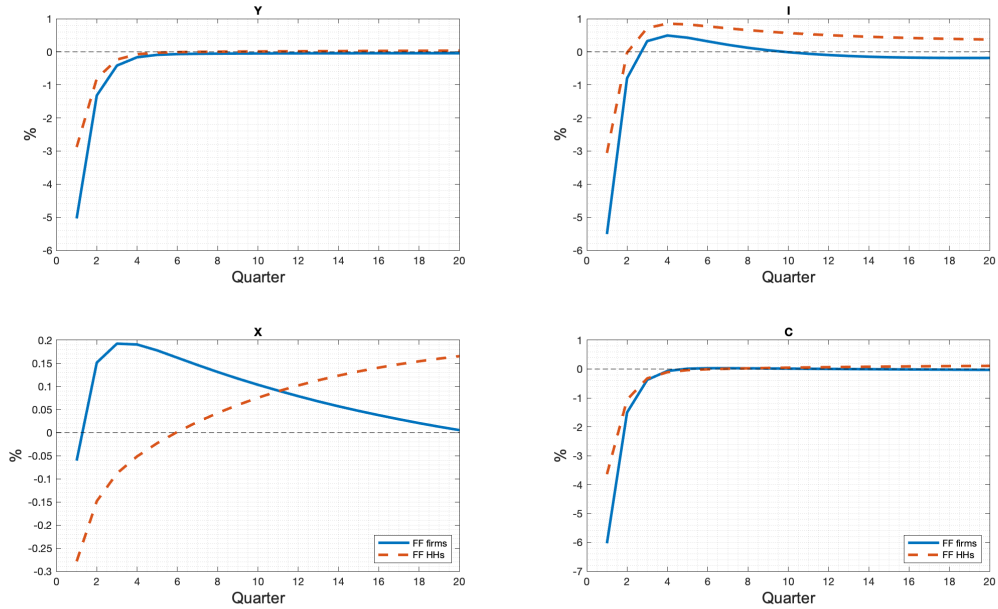

Figure D.1: Impulse response to a monetary contraction for aggregate variables

Note: monetary shock is $\epsilon^R = 0.0025$ in both the scenarios. The blue line refers to an economy with financial frictions on firms, the red one when frictions are on households.

D Impulse responses of MP contractionary shock - Same shock magnitude

Below, I show the main aggregate and inequality fluctuations when the monetary shock magnitude used to produce IRFs is the same in both scenarios. Results are qualitatively similar to the findings in the baseline model, where I consider two different shock magnitudes that have the same effect on output.

E Consumption inequality analysis for goods consumption C

In this section, I show the fluctuations in the Gini index and share averages for total goods consumption, C, for the baseline model. Also in this case, when financial frictions on households are active, the changes in the Gini index are stronger, as show in Figure E.1. This can be explained by Figure E.2: while fluctuations of aggregate C are similar in the two scenarios, average consumption for top e bottom shares of households are more scattered in the case of financial frictions on households.

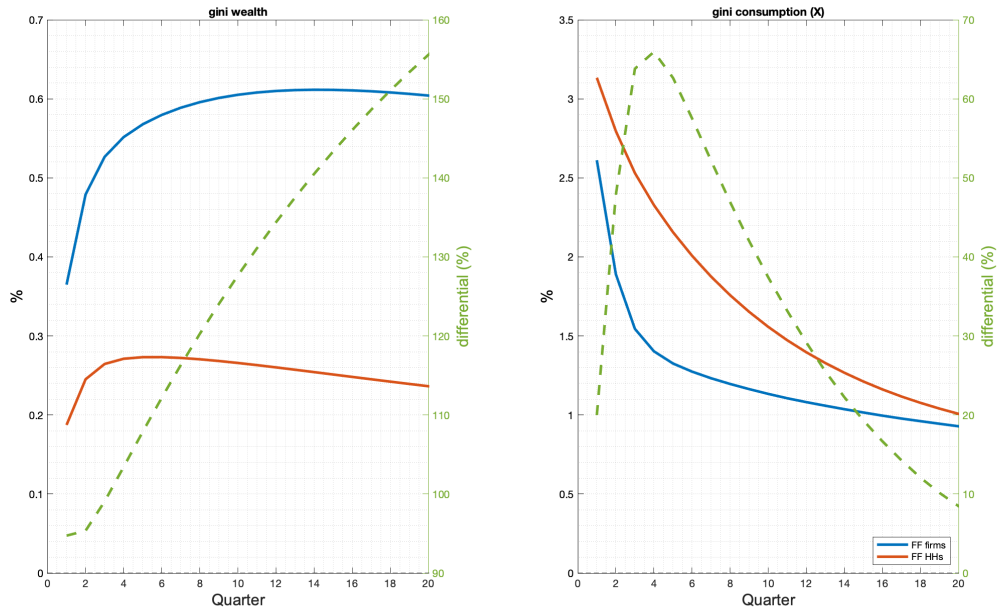

Figure D.2: Impulse responses to a monetary contraction for wealth and consumption inequality.

Note: monetary shock $\epsilon^R = 0.0025$. The blue line refers to an economy with financial frictions on firms, the red one when frictions are on households. The green dotted line is the percentage difference between the two curves using the lower curve as the base.

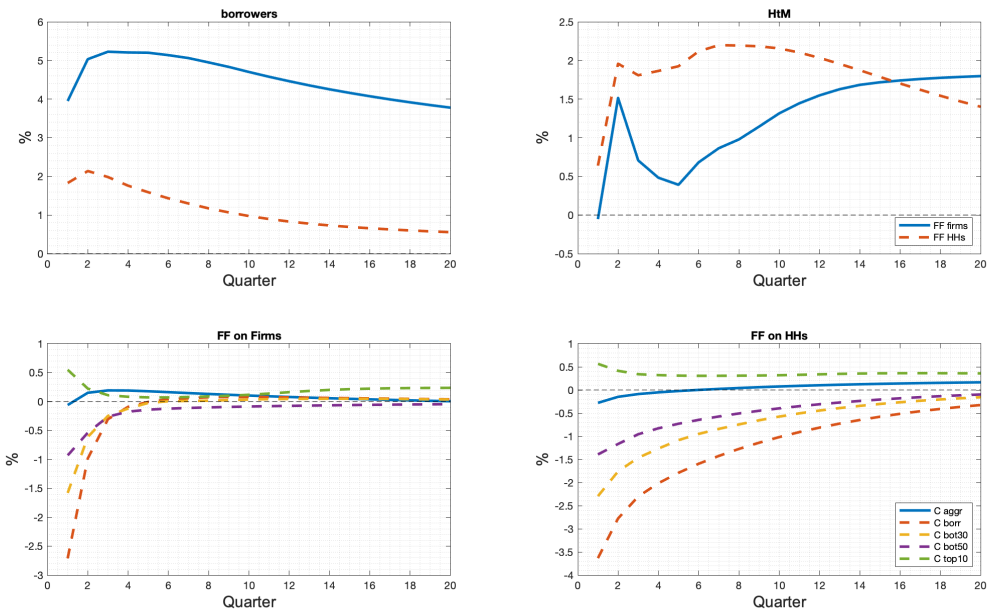

Figure D.3: IRFs for wealth and consumption decomposition

Note: monetary shock $\epsilon^R = 0.0025$. The top of the figure shows the wealth analysis for households around the zero-wealth threshold, while the bottom shows the average consumption decomposition for composite consumption.

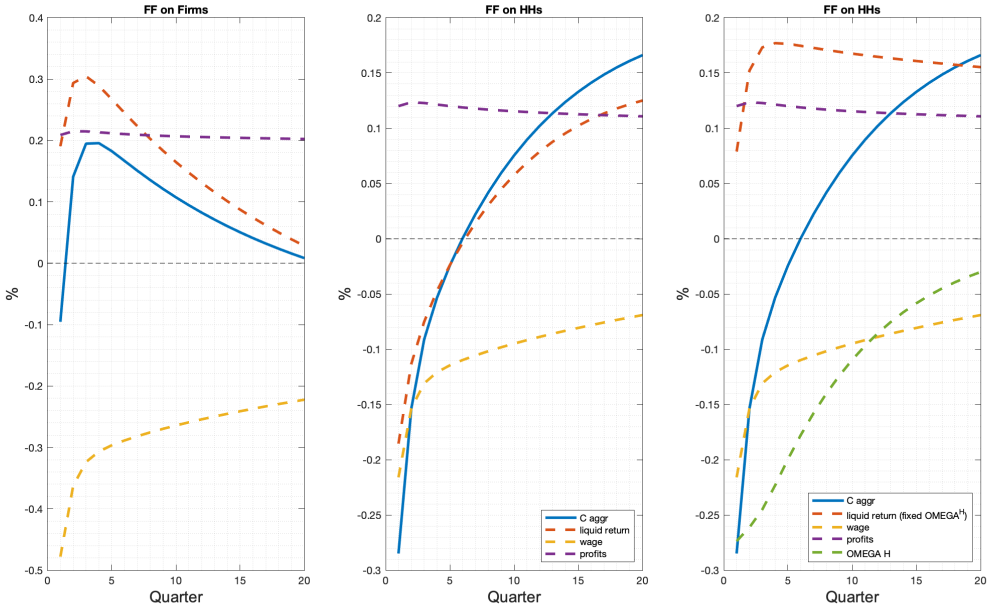

Figure D.4: Consumption decomposition for relevant prices

Note: monetary shock $\epsilon^R = 0.0025$. The graph on the left-hand side represents the decomposition for the case of frictions on firms and the other two represent the case of frictions on household borrowing. In the graph on the right-hand side, I consider the borrowing penalty ω_t^H as an individual variable.

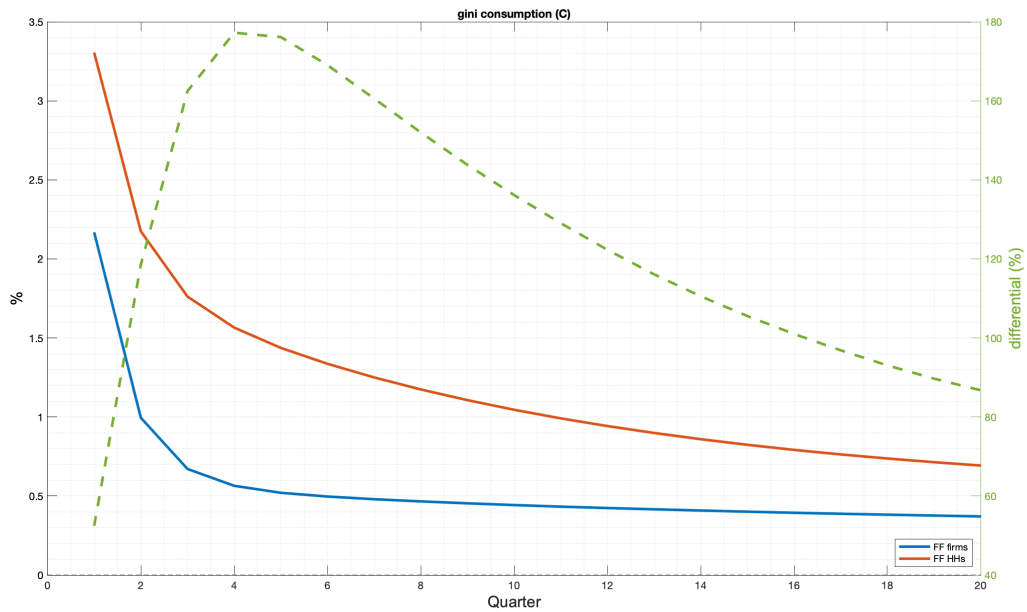

Figure E.1: Impulse responses to a monetary contraction for consumption inequality.

Note: monetary shock $\epsilon^R = 0.0025$ for active financial frictions on household borrowing, $\epsilon^R = 0.0014$ otherwise. The blue line refers to an economy with financial frictions on firms, the red one when frictions are on households. The green dotted line is the percentage difference between the two curves using the lower curve as the base.

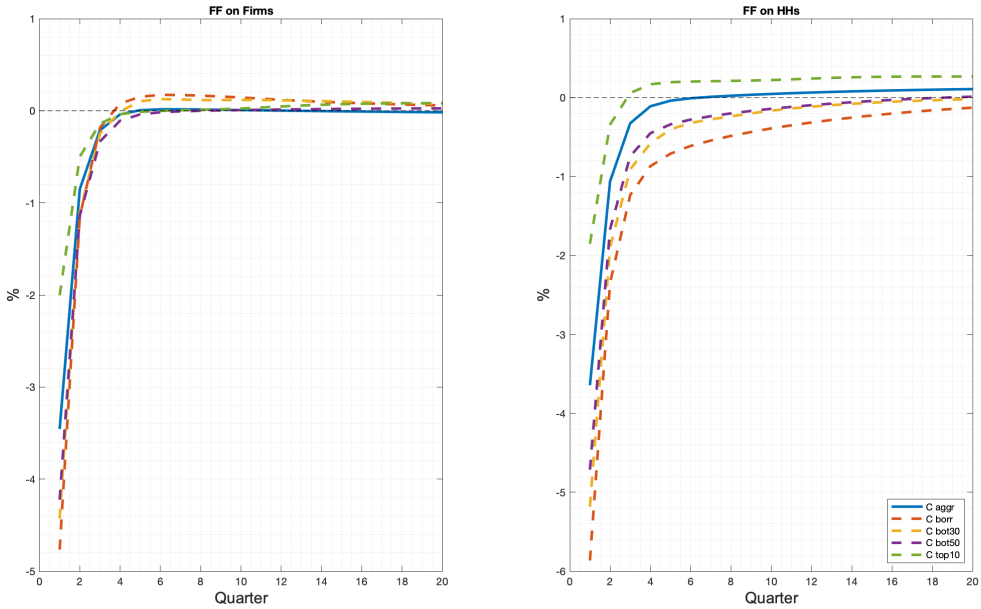

Figure E.2: Average consumption fluctuation for different shares of households.
Note: monetary shock $\epsilon^R = 0.0025$ for active financial frictions on household borrowing, $\epsilon^R = 0.0014$ otherwise.

F Robustness checks

In this section, I show the Gini indices and consumption decomposition according to prices for different variants of the baseline model. Figure F.1 and Figure F.2 show results when the parameter regulating the fiscal policy, ρ_{gov}, is equal to zero. Since I employ a HANK model, the Ricardian equivalence does not hold, and changes in fiscal policies could have significant effects. Figure F.3 and Figure F.4 display results for the case limit of no quadratic costs for capital producer, that is, $\phi = 0$. I also consider a model in which I change the parameter for households' risk aversion, ξ. In the baseline calibrations, I assume $\xi = 4$ as in Bayer et al. (2019), but other models in the HANK literature(e.g., Auclert et al., 2021), assume a lower risk aversion for households. Therefore, in Figure F.5 and Figure F.6, I present the results when assuming a model with $\xi = 2$. In this case, however, to match wealth distribution moments, I need to change other parameters, such as β, ζ and $\underline{a}$.

The main findings of the baseline model, that is, relatively higher wealth inequality for financial frictions on firms, relatively higher consumption inequality for financial frictions on households, and the relevance of the borrowing penalty ω^H for this dynamic, are robust to these changes in parameterization.

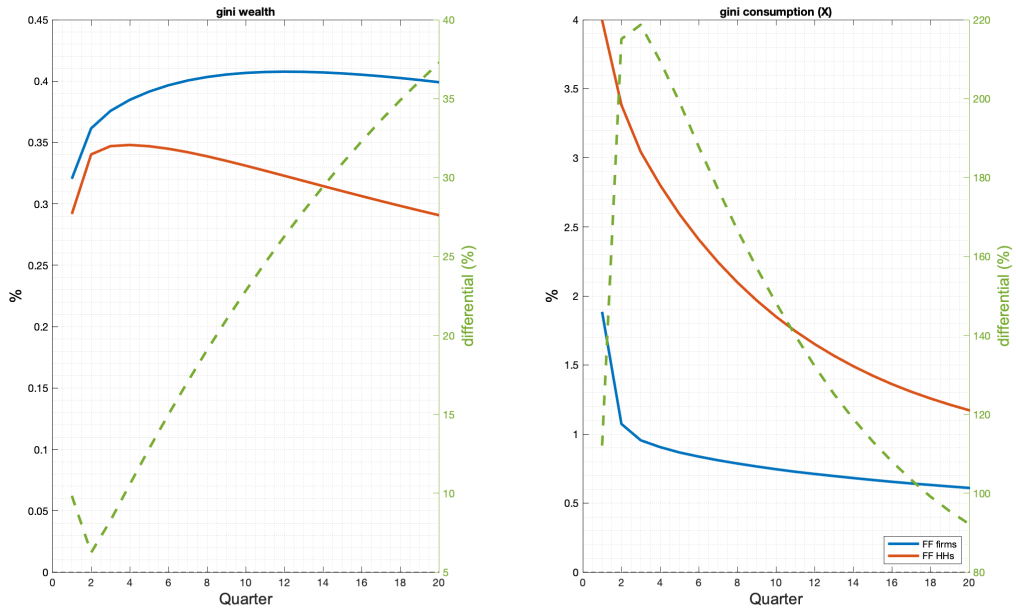

Figure F.1: Impulse responses to a monetary contraction for wealth and consumption inequality, $\rho_{gov} = 0$.

Note: monetary shock $\epsilon^R = 0.0025$ for active financial frictions on household borrowing, $\epsilon^R = 0.0014$ otherwise. The blue line refers to an economy with financial frictions on firms, the red one when frictions are on households. The green dotted line is the percentage difference between the two curves using the lower curve as the base.

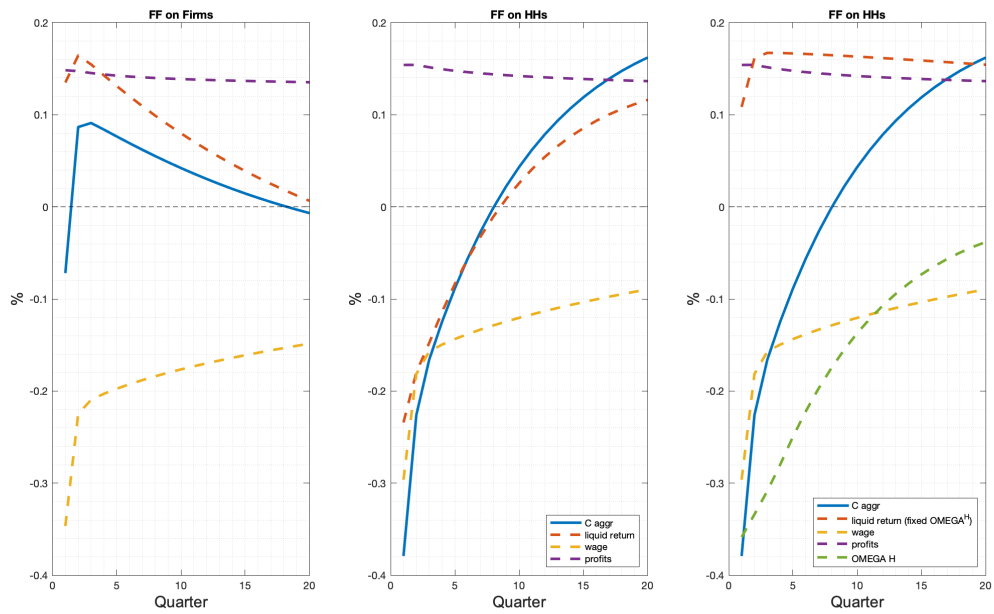

Figure F.2: Consumption decomposition for relevant prices, $\rho_{gov} = 0$

Note: monetary shock $\epsilon^R = 0.0025$. The graph on the left-hand side represents the decomposition for the case of frictions on firms and the other two represent the case of frictions on household borrowing. In the graph on the right-hand side, I consider the borrowing penalty ω_t^H as an individual variable.

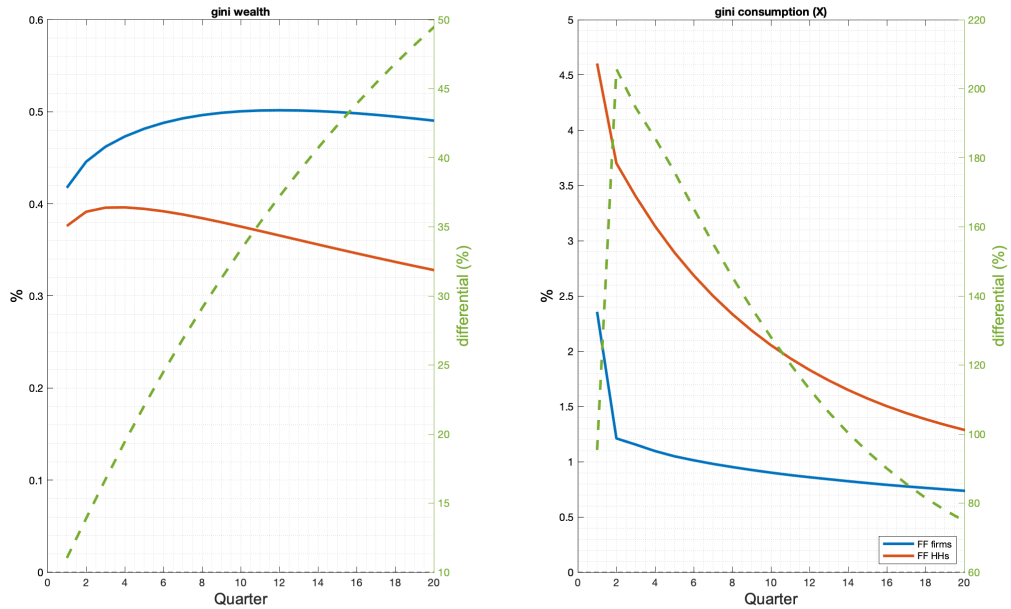

Figure F.3: Impulse responses to a monetary contraction for wealth and consumption inequality, $\phi = 0$.

Note: monetary shock $\epsilon^R = 0.0025$ for active financial frictions on household borrowing, $\epsilon^R = 0.0014$ otherwise. The blue line refers to an economy with financial frictions on firms, the red one when frictions are on households. The green dotted line is the percentage difference between the two curves using the lower curve as the base.

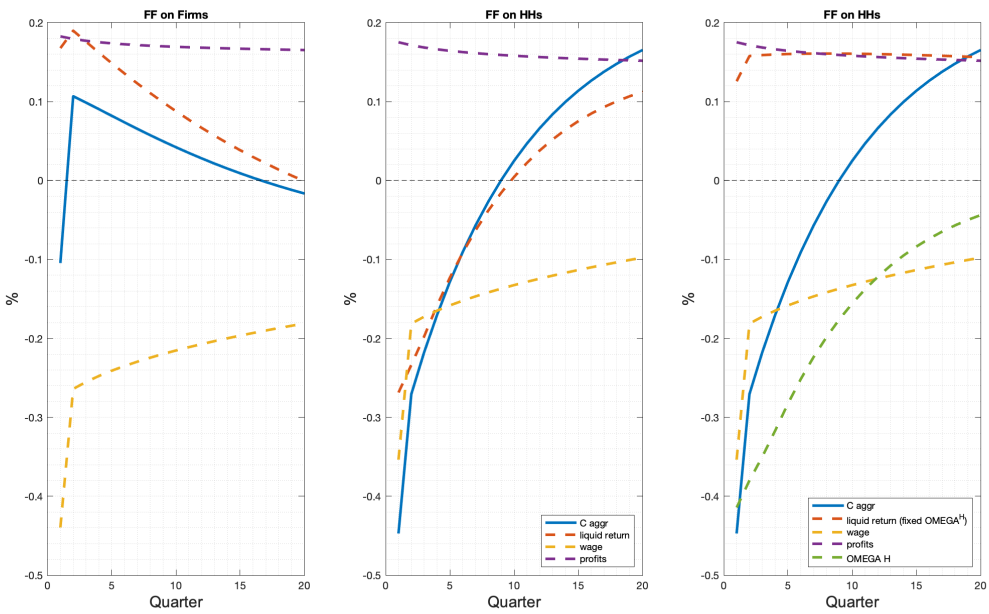

Figure F.4: Consumption decomposition for relevant prices, $\phi = 0$

Note: monetary shock $\epsilon^R = 0.0025$. The graph on the left-hand side represents the decomposition for the case of frictions on firms and the other two represent the case of frictions on household borrowing. In the graph on the right-hand side, I consider the borrowing penalty ω_t^H as an individual variable.

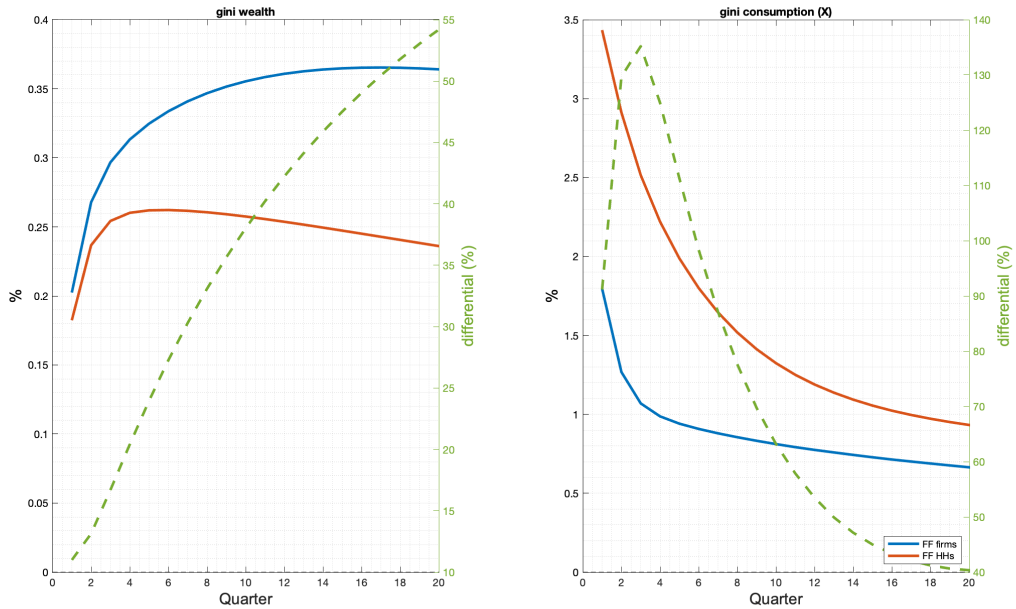

Figure F.5: Impulse responses to a monetary contraction for wealth and consumption inequality, $\xi = 2$.

Note: monetary shock $\epsilon^R = 0.0025$ for active financial frictions on household borrowing, $\epsilon^R = 0.0014$ otherwise. The blue line refers to an economy with financial frictions on firms, the red one when frictions are on households. The green dotted line is the percentage difference between the two curves using the lower curve as the base.

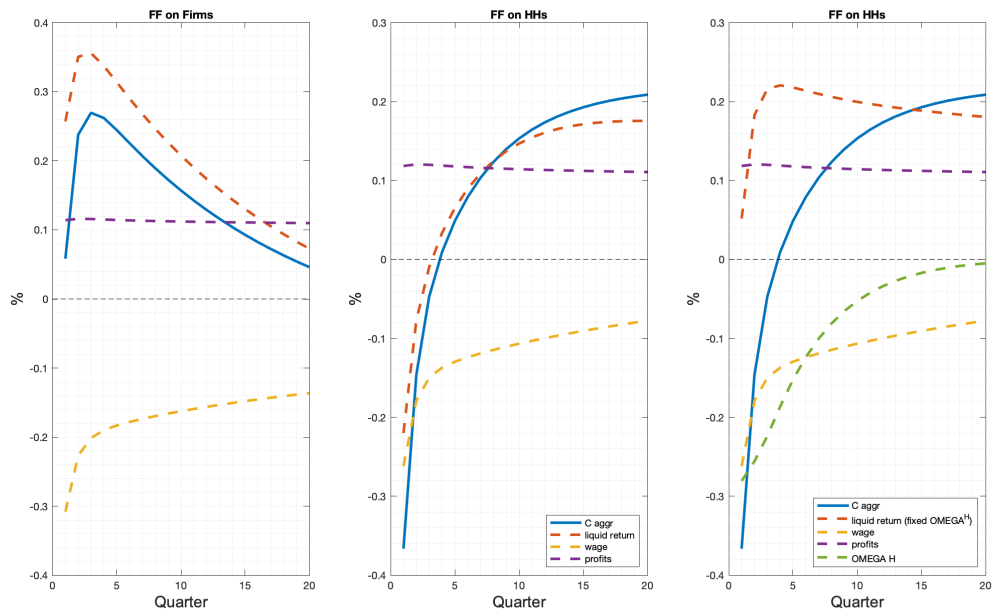

Figure F.6: Consumption decomposition for relevant prices, $\xi = 2$

Note: monetary shock $\epsilon^R = 0.0025$. The graph on the left-hand side represents the decomposition for the case of frictions on firms and the other two represent the case of frictions on household borrowing. In the graph on the right-hand side, I consider the borrowing penalty ω_t^H as an individual variable.